Master Thesis

Automated Multimodal Fact-Checking with an MLLM-Based Agentic System Evaluated on Two New Datasets: The Gaza and Ukraine Wars

M.Sc. in Social Data Science

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Abstract

This thesis evaluates DEFAME (Dynamic Evidence-based FAct-checking with Multimodal Experts), a zero-shot MLLM-based agentic fact-checking system, on two newly created datasets that address misinformation in high-stake contexts: the wars in Gaza and Ukraine. The Gaza-Israel and Ukraine-Russia datasets contain four different claim types (text-only claims, claims with normal images, AI-generated and altered images) and recent claims from June 2024 to April 2025. Extensive experiments with five different DEFAME variants, each replicated three times, revealed a robust performance of DEFAME across all four claim types on both datasets. Whereas claims with altered images were the most difficult to verify on the Gaza-Israel dataset, it was text-only claims on the Ukraine-Russia dataset. Ablation studies showed that removing the web search tool was associated with the highest drop in performance, particularly for text-only claims. Removing reverse image search only moderately affected claims with normal images on the Gaza-Israel dataset, but did not reveal performance changes for claims with AI-generated or altered images. The qualitative analysis yielded novel failure reasons of DEFAME: (1) missing verification of altered images, (2) misinformation and/or conflicting evidence in new sources and (3) the context of retrieved evidence making false claims appear plausible. The study further revealed reliability problems in ground truth labels from professional fact-checking websites. Future work involves comparing DEFAME to more recent agentic systems on the new datasets and applying DEFAME to other modalities that contribute to the spread of misinformation in wars, e.g. video and audio data.

Contents

List of Tables

List of Figures

1 Introduction

Modern wars are also information wars: misinformation about the enemy is deliberately created, spread and reinforced on (social) media platforms. Misinformation campaigns create narratives and discourses to shape public opinion and to legitimize war crimes. Examples in current times are claims that Ukraine started the war with Russia or that Palestinians use crisis actors to fake their deaths in Gaza ("Pallywood" allegations). With the increase of social media users worldwide and AI-generated content, the scale of misinformation grows beyond the capacity of professional human fact-checkers. Thus, automated approaches are necessary.

Recent advances in the research of large language models (LLMs) have introduced autonomous agents which are able to achieve a goal by planning, executing, summarizing and/or reasoning about actions on their own (with the use of external tools) [65, 63, 42, 26]. This has inspired researchers to apply LLM-based agentic systems to automatic fact-checking. One recent example of such a system is DEFAME (Dynamic Evidence-based FAct-checking with Multimodal Experts), which is a modular, zero-shot MLLM pipeline to verify multimodal claims [1]. It has achieved State-of-the-Art (SOTA) results on existing multimodal datasets and is open-source. Thus, it will be used in this thesis.

Yet, existing multimodal datasets cover general (political) topics, do not contain new modalities, such as AI-generated and altered images, and develop the data leakage problem over time: their claims might become part of the pre-training data of LLMs, which compromises the verification task. Datasets about the wars in Ukraine and Gaza, which cover new modalities and recent claims do not exist.

Therefore, this thesis contributes to the literature by:

1. creating the Gaza-Israel dataset and Ukraine-Russia dataset that cover four claim types (text-only claims, claims with normal images, claims with AI-generated images and claims with altered images) and recent claims from June 2024 - April 2025

2. evaluating DEFAME on the two new datasets with a two-level approach: a dataset

and claim-type level

3. revealing reliability problems in the ground truth labels of existing professional fact-checking websites

4. and identifying new failure reasons of DEFAME

To guide the evaluation of DEFAME on the two new datasets, the following research questions are posed:

1. *How does DEFAME perform on the new Gaza-Israel and Ukraine-Russia dataset?*

2. *How does DEFAME perform across the four claim types, i.e. text-only claims, claims with normal images, AI images and altered images?*

3. *How does DEFAME's performance vary across the four claim types when individual tools (web search, reverse image search, image search, and geolocation) are removed?*

4. *What are DEFAME's failure reasons on the new datasets?*

The remainder of this thesis is structured as follows. First, the general automated fact-checking process is explained (2.1), the new datasets and DEFAME are situated within existing multimodal datasets (2.2) and MLLM-based fact-checking approaches (2.3) and the modular architecture of DEFAME is described (2.4).

Second, the steps of the data pipeline to create the new datasets are described (3.1, 3.2), the reliability problems of fact-checking websites are discussed (3.3), the final descriptive statistics of the new datasets (3.4) and the ethical and legal context of the data collection (3.5) are presented.

Third, the evaluation of DEFAME is presented. The experimental setup and evaluation strategy is explained (4.1), the results on the Gaza-Israel dataset (4.2) and Ukraine-Russia dataset (4.3) are presented and the main results and its limitations are discussed (4.4).

Last, the conclusion (5) summarizes the main results of the thesis and points out future work on the new datasets and DEFAME that would contribute to existing research of automated multimodal fact-checking.

2 Literature Review

2.1 Automated Multimodal Fact-Checking

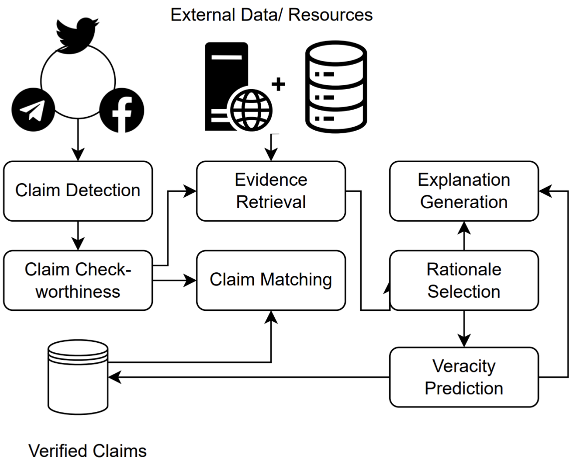

Figure 1: A typical claim verification pipeline. Source: [11]

Figure 1 visualizes the steps of a typical claim verification pipeline in automated fact-checking (AFC). The pipeline consists of seven steps, yet four of the steps are commonly mentioned in most of the AFC literature: claim detection, evidence retrieval, veracity prediction and explanation/justification generation [27].

Step 1, claim detection, is designed to identify claims in a larger corpora of text [11]. This step is not relevant for my datasets, since the fact-checking websites already perform this step. Step 2, claim check-worthiness, identifies relevant claims that are check-worthy [11]. In my data pipeline to create the datasets, which is shown in Chapter 3 in Figure 3, claim check worthiness is performed in the manual data collection through the third step *selection of articles* and in the API data collection through the pre-processing pipeline. All irrelevant claims that do not match my criteria are filtered out. The detailed steps and the criteria are described in Chapters 3.1.1 and 3.1.2. In step 3, claim matching, the claims are matched to already fact-checked claims [11]. By accessing databases it can be checked, whether the claim was fact-checked before. This can help to directly extract the veracity label. In step 4, evidence retrieval, information in the form of text, image, video or trabular data is retrieved from claim-independent sources which is used to predict the veracity of a claim [11, 27].

Step 5, rationale selection, is performed to select the relevant out of all retrieved information [11]. Not all information is relevant to predict the veracity of a claim. In step 6, veracity prediction, a classifier predicts the label of a claim. The input is the claim, an image or a video, and the label definitions. Typical claim labels are *Supported*, *Refuted*, *Misleading* and *Not Enough Information (NEI)* [11, 27]. In the last step 7, explanation/justification generation, the used model or system needs to justify its verdict [11, 27]. This increases the transparency, readability and explainability of the fact-check and is particularly helpful when LLMs are employed to decode their block box nature.

Key challenges that occur in AFC during one or multiple of the presented steps above are: claim ambiguity [44], the quality and relevance of retrieved evidence [27], multilinguality [27, 11], pipeline scalability [69], synthetic datasets [69], imbalanced datasets [69], and label choice (binary vs. multi-class classification task) [27]. Overall, the task of AFC must be differentiated from adjacent tasks such as fake news detection, rumor detection, clickbait detection and commonsense reasoning [69].

2.2 Multimodal Fact-Checking Datasets

Table 1 shows an overview of existing and my two new multimodal fact-checking datasets. The table only shows datasets that are (1) multimodal (text-image, text-video), (2) that have been used to evaluate DEFAME in the original paper (MOCHEG, VERITE and CLAIMREVIEW2024+) [1] or (3) that are topically related to my datasets (WarClaim). The table displays their names, the paper publication date, the number of claims, the claims' source and time frame, the topics covered, the claim types and whether a data leakage problem exists or not.

Dataset Name	Paper Date	Claims (n)	Source	Claims Date	Topic(s)	Claim Types	Data Leakage
MOCHEG (Multimodal Fact-Checking and Explanation Generation) [68]	07/2023	15.601	Snopes, PolitiFact	N/A	General	Text-Only, Normal Images	✓
VERITE (VERification of Image-TExt pairs) [50]	01/2024	1.001	Reuters, Snopes	01/2021 - 01/2023	General	Normal Images	✓
WarClaim [57]	06/2024	5.000	131 fact-checking websites	10/2023 - 02/2024	Specific (Gaza-Israel)	Videos	✓
CLAIMREVIEW2024+ [1]	12/2024	300	Google FactCheck Claim Search API	11/2023 - 01/2025	General	Text-Only, Normal Images	✓
Gaza-Israel Dataset	09/2025	100	AFP, Reuters, Snopes, PolitiFact, Google FactCheck Claim Search API	07/2024 - 04/2025	Specific (Gaza-Israel)	Text-Only, Normal Images, AI Images, Altered Images	×
Ukraine-Russia Dataset	09/2025	79	AFP, Reuters, Snopes, PolitiFact, Google FactCheck Claim Search API	07/2024 - 04/2025	Specific (Ukraine-Russia)	Text-Only, Normal Images, Altered Images	×

Table 1: An overview of existing and the two multimodal fact-checking datasets proposed in this paper. The table shows the datasets' names, publication date, size, the sources used, claims' date, the topics and claim types covered in the datasets and whether a data leakage problem exists.

Note that the existence of a data leakage problem is always dependent on the MLLM that is used as a backbone within DEFAME. I'm using Gemini 2.0 Flash-Lite in my experiments, see the reasons in Chapter 2.4, which has a knowledge cutoff in June 2024 [18]. Thus, all datasets that contain claims up to June 2024 have a data leakage problem, i.e. all datasets shown in Table 1 except for my new created datasets. Since the datasets use public data to extract claims, mostly social media platforms, these claims are most likely part of the training data of current or future MLLMs [1]. As a result, it is difficult to attribute a prediction to either the pre-trained parametric knowledge of the MLLM or the multimodal evidence retrieval. This might compromise the evaluation of an automated multimodal fact-checking system.

To test this assumption, the authors of DEFAME have created the CLAIMREVIEW2024+ dataset which contains claims that are scraped *after* the knowledge cutoff of their MLLM used (GPT-4o; October 2023). Their experiments show that DEFAME outperforms the two MLLM-only baselines which do not use any retrieval, GPT-4o and GPT-4o Chain-of-Thought (CoT), with a significantly higher degree on CLAIMREVIEW2024+ than on MOCHEG and VERITE [1]. This indicates that the latter two have most likely been part of the training data of GPT-4o and that external tools for multimodal retrieval is especially relevant for fact-checking recent claims, i.e., claims with dates *after* the knowledge cutoff of an MLLM.

For this reason, the topically related WarClaim dataset, which contains claims about the Gaza-Israel war, is insufficient since the covered claims only reach to February 2024 and thus the data leakage problem exists. Further, the dataset only contains one modality: videos.

This thesis contributes to the existing literature by creating two new multimodal fact-checking datasets that focus on two specific high-stake wars (Gaza, Ukraine) where a lot of misinformation exists. In contrast to existing work, the datasets contain two new modalities (AI-generated and altered images) and cover recent claims from July 2024 to April 2025. The overall pipeline and the concrete steps to create the datasets are described in Chapter 3.

2.3 (M)LLM-based Automated Fact-Checking

The literature review focuses on (M)LLM-based approaches for automated fact-checking. The reasons are that these methods achieve SOTA results on existing datasets and that DEFAME also uses an MLLM as its backbone. Other approaches that, e.g., utilize graph structures [13, 49], are not considered here.

This thesis categorizes the multitude of LLM-based AFC systems similarly to [11, 71] in fine-tuning and prompt-based approaches. The prompt-based approaches are divided in two sub-categories: static prompting and agentic systems. The focus will be on agentic AFC systems, since DEFAME is also one.

2.3.1 Fine-Tuning Approaches

In the fine-tuning paradigm a LLM updates its pre-trained weights using labeled data from a downstream task [9, 52]. In this context fact-checking is the downstream task. Since most of the approaches focus on text-only benchmarks, they will only be briefly mentioned. For example [4] fine-tunes BART [38] on the FEVER [59] benchmark, whereas [5] introduce a framework that splits fact-checking into a question-answering and a text classification task where the authors combine a FakeNet model and fine-tuned LLMs. Other approaches use reinforcement learning to fine-tune models for improving the veracity of claims and supporting evidence [70, 29].

2.3.2 Prompting Approaches

Static Prompting: In-Context Learning (ICL)

In-Context Learning (ICL) refers to the capability of LLMs of learning a task by providing a few examples of context and completion (*few-shot prompting*, [3]) or describing the task with natural language (*zero-shot prompting*, [36]) in a prompt. To improve the reasoning abilities of (M)LLMs *Chain-of-Thought (CoT)* prompting has been introduced [64]. Here, the MLLM is prompted to think step by step and break down the task in manageable sub-tasks. In contrast to fine-tuning, the pre-trained model weights are not updated in prompt-based approaches and the need for labeled task-specific data is greatly reduced.

In the fact-checking context, (M)LLMs are provided with the input claim and modality (e.g, an image or video) and either a description of the verification task (*zero-shot prompting*) or with some examples of the verification process. Thus, the MLLM relies on its parametric knowledge and the prompt to predict the truthfulness of a claim. Examples are the GPT-4o and GPT-4o CoT models evaluated in the original DEFAME paper, which showed remarkable results on the MOCHEG and VERITE datasets, but performed significantly worse on the CLAIMREVIEW2024+ datasets with claims after their knowledge cutoff.

Agent-based Approaches

Although ICL approaches are powerful, they still have major problems: LLMs are prone to hallucination, their parametric knowledge is static and their predictions lack links to verifiable sources [1].

To reduce hallucinations of LLMs and ground their answers in external sources, the technique of Retrieval-Augmented Generation (*RAG*, [39]) has been introduced. In the context of AFC, external evidence can be retrieved to get more information about a claim. Most claims of the existing multimodal fact-checking datasets are public claims that are discussed on social media platforms or in newspapers. Therefore, evidence retrieval is performed through external tools, such as web search, image search or reverse image search.

The tool use brings us to the next innovation in the field of LLM research: LLM-based autonomous *agents*. Even though this new concept lacks a shared and consistent definition in the literature, LLM-based agents have common key characteristics: they autonomously achieve a goal by using an (M)LLM as their core decison-making component and utilize some or all of the following capabilities: planning, (external) tools, reasoning, self-reflection and iterative action execution [63, 65, 58, 42, 26, 41].

Table 2 provides an overview of the most relevant and published (M)LLM-based agentic systems in the AFC context. The AFC systems are ordered by their publication date. The table lists their capabilities to process multimodal claims and evidence, use evidence retrieval, planning, tools, reasoning and to produce an explainable answer. It also shows the backbone of the system: an LLM or MLLM.

From the available AFC systems in the literature only those are classified as agents in this thesis, that support at least two of the following three capabilities: planning, tools, and reasoning. This is the case for each of the AFC systems listed in Table 2.

When DEFAME was published its main innovation was to combine the fragmented AFC landscape into one *end-to-end solution* that can process multimodal claims and evidence, retrieve evidence through multiple tools, employ planning and reasoning and return a detailed fact-check report for each verified claim [1]. Only one other AFC system that is listed in Table 2 and published after DEFAME also meets all of the conditions: Multi-Agent Deep Research. At this point, the decision to use DEFAME was already made.

AFC System	Publication date	Multimodal claims	Multimodal evidence	Evidence retrieval	Planning	Tools	Reasoning	Explainable	Backbone
ProgramFC [48]	05/2023	×	×	✓	✓	✓	◑	◑	LLM
SELF-CHECKER [40]	06/2024	×	×	✓	✓	×	✓	◑	LLM
PACAR [71]	05/2024	×	×	✓	✓	✓	✓	✓	LLM
RAGAR [35]	07/2024	✓	×	✓	×	✓	✓	✓	MLLM
DEFAME [1]	12/2024	✓	✓	✓	✓	✓	✓	✓	MLLM
MDAM^3 [67]	04/2025	✓	✓	✓	×	✓	×	✓	MLLM
Multi-Agent Deep Research [37]	06/2025	✓	✓	✓	✓	✓	✓	✓	MLLM
T^2 Agent [6]	06/2025	✓	✓	✓	✓	✓	✓	×	MLLM
CRAVE [10]	07/2025	✓	✓	✓	×	✓	✓	✓	MLLM
MultiReflect [34]	08/2025	✓	✓	✓	×	✓	✓	✓	MLLM

Table 2: An overview of the most relevant and published (M)LLM-based agentic AFC systems. The table structure is inspired [1]. The AFC systems are ordered by publication date. The table lists the capabilities of an AFC system: to process multimodal claims and evidence, to use evidence retrieval, planning, tools, reasoning and to produce an explainable answer. Lastly, it shows whether an LLM or MLLM is used as its backbone. The symbols ✓, ◑ and × denote that a capability is either fully, half or not supported.

Beyond meeting all the requirements listed in Table 2, the modular and open-source nature of DEFAME and its SOTA results on existing multimodal datasets such as MOCHEG and VERITE were the major reasons to use it. Its modular, six-step pipeline will be described next.

2.4 DEFAME: Dynamic Evidence-based FAct-checking with Multimodal Experts

Figure 2 visualizes DEFAME's six-stage pipeline which involves three main components: an MLLM, a fact-checking report, and external tools. Five of the six stages involve MLLM prompting. DEFAME breaks down the fact-checking process into six stages, mimicking real-world human fact-checkers [1].

The stages are shortly described here and can be read in detail in the original paper [1]. In stage 1, *plan actions*, the MLLM receives the claim and the image as input and is prompted to propose an action sequence to retrieve missing evidence. In-context learning

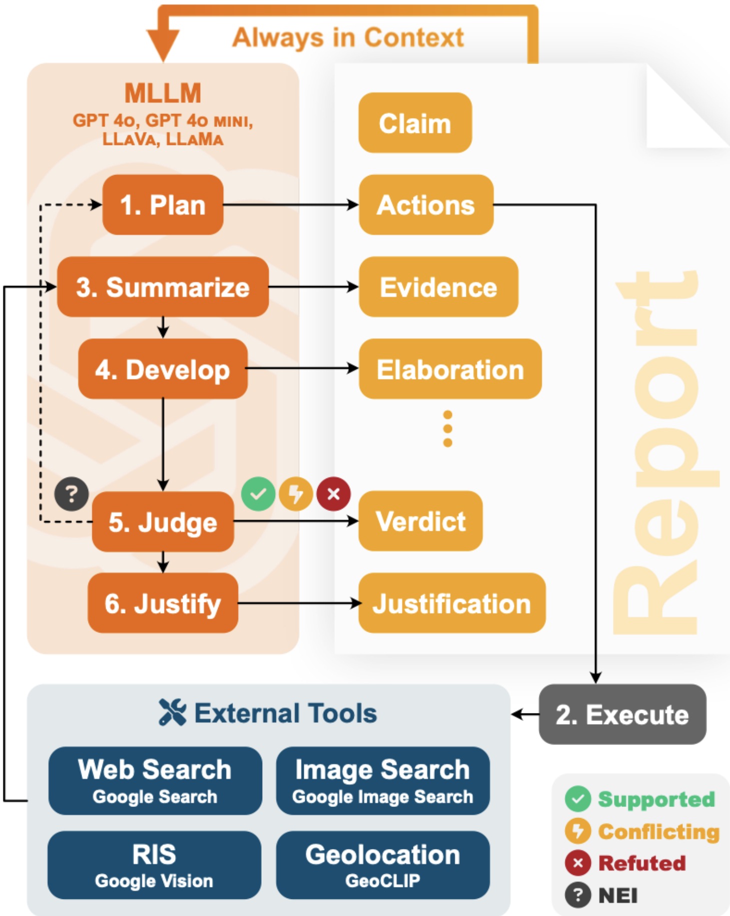

Figure 2: An overview of DEFAME's six-stage pipeline with its three main components: an MLLM, a fact-checking report that is produced and external tools. Source: [1]

(ICL) helps the module to decide which of the available tools to use: web search, image search, reverse image search or geolocation. For text-only claims only web and image search can be performed, while for claims with images all tools can be invoked. The text queries for web and image search are generated dynamically.

In stage 2, *execute actions*, DEFAME invokes the tools based on the actions planned in the previous stage. For the web search, a textual search query is used to provide the top three relevant web pages. By default, Google Search via the Serper API is used [56]. For the image search, up to three URLs of web pages containing images, diagrams and infographics are returned given a textual caption. By default Google Image Search via the Serper API is used. For the reverse image search, the Google Vision API [22] is used to retrieve up to three URLs of web pages including the same image the claim is referencing. For geolocation, the most likely countries an image originates from is estimated using GeoCLIP [62]. The page content of each returned URL is scraped with Firecrawl [16]. The fact-checking websites which are used to collect the claims and their labels are excluded to ensure a meaningful fact-checking setting. A list of the excluded fact-checking websites and excluded domains due to bot traffic restrictions is provided in Appendix B in Tables 14 and 15.

In stage 3, *summarize results*, the MLLM is prompted to reason about the retrieved evidence and generate a summary of the key findings for each tool output.

In stage 4, *develop the fact-check*, DEFAME compares the claim with the summarized evidence. The MLLM is prompted to discuss the claim's veracity step-by-step.

In stage 5, *judge/predict a verdict*, DEFAME predicts the label of the claim. The MLLM is prompted to summarize the key findings from the previous stages before making the prediction. If the retrieved evidence is insufficient to verify a claim, DEFAME returns *Not Enough Information (NEI)* and returns to stage 1 of the pipeline to collect more evidence. DEFAME dynamically adjusts the text queries for web and image search in subsequent rounds of retrieval. The maximum number of iteration rounds is set to three.

In stage 6, *justify the verdict*, the MLLM is prompted to justify its verdict by creating a short summary with the key findings and evidence from the fact-checking process.

The final output of DEFAME is a pdf report that documents the fact-checking process for each claim. This greatly increases the explainability, human readability and transparency of the automated fact-checking procedure. It helps to understand how DEFAME came to its final decision. An example is provided at the end of Appendix B.

With regards to the MLLM used as a backbone of DEFAME, I decided to use Google's Gemini 2.0 Flash-Lite model. The default MLLM of DEFAME, OpenAI's GPT-4o (mini), could not be used due to restrictive rate limits in the low usage tiers (Free, Tier 1) [47]. In constrast, the rate limits of Google's Gemini models are more permissive in the lower usage tiers [21] and enable the processing of all the evidence retrieved and generated by

DEFAME. Gemini 2.0 Flash-Lite was used due to its cost efficiency and low latency [19]. Its knowledge cutoff date (June 2024) determined the time frame of the creation of the datasets, which is described next.

3 Own Multimodal Fact-Checking Datasets: Gaza and Ukraine War

This chapter describes the pipeline to create the new Gaza-Israel and Ukraine-Russia datasets and presents their descriptive statistics. Figure 3 gives a high-level overview of the pipeline with its two pillars: the manual and the API data collection. The time frame for collecting claims has been set to be from July 1st, 2024 to April, 30th 2025. Thereby, the claims *postdate* the knowledge cutoff date (June 2024) of the MLLM (Gemini 2.0 Flash-Lite) used in the experiments in Chapter 4 and the problem of *data leakage* is minimized.

I used both manual and API data collection because only using manual collection would have resulted in datasets that are too small, and only using API data collection would have resulted in incomplete datasets: Google's FactCheck Claim Search API [23] did either not include any claims (Reuters) or not all claims (AFP Factcheck, Snopes, PolitiFact) for the fact-checking websites used in the manual data collection in the specified time frame.

The remaining chapter is structured as follows: First, I describe the steps of the manual and API data collection (3.1). Next, I describe the post-processing steps after the data have been collected (3.2). Then I share the results of preliminary DEFAME experiments on my datasets, which revealed wrong ground truth labels of the fact-checking websites and the strategy I employed to deal with the problem (3.3). Finally, I present the descriptive statistics of both datasets to give an overview of the label and claim type distributions (3.4).

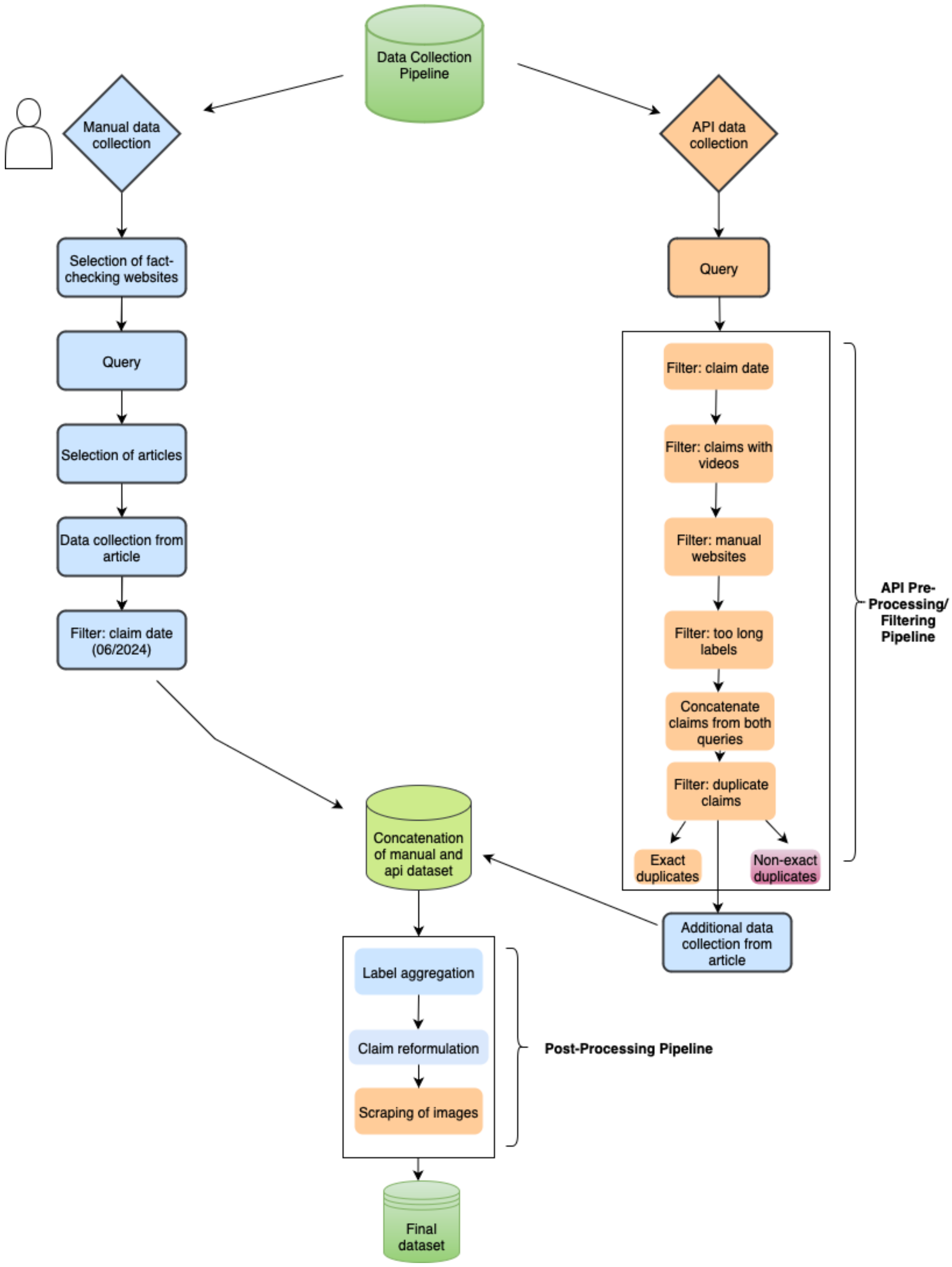

Figure 3: A high-level overview of the pipeline to create the Gaza-Israel and the Ukraine-Russia dataset. The database symbol represents a dataset, the diamond symbol the data collection type and the rectangular form represents one step in the pipeline. The blue color denotes a manual step, the orange color a computational step and the purple color a hybrid step with manual and computational parts. The green color denotes a dataset. Source: Own creation.

3.1 Data Collection

3.1.1 Manual Data Collection

As displayed in Figure 3 the first step was to select professional, reliable and independent fact-checking websites. I decided to collect claims from AFP Factcheck, Reuters, Snopes and PolitiFact. The latter three websites were also used to create previous datasets, as shown in Table 1, and all four websites are members of the International Fact-Checking Network (IFCN)[32] and active verified signatories of the IFCN's Code of Principles[31]. Each website is subject to the IFCN's vetting process and is evaluated by external assessors. The verification is valid for one or two years. As of the date of writing, the verification of all four used fact-checking websites is still valid.

Website	**Query**	**Query Type**	**URL**
AFP Factcheck	*Israeli-Palestinian conflict*	Website	Category
Reuters	*Gaza Fact Check*, *Israel Fact Check*	Own Query	Gaza Query, Israel Query
Snopes	*Gaza*, *Israel*	Own Query	Gaza Query, Israel Query
Politifact	*Israel*	Website	Category

Table 3: An overview of the queries used for the manual data collection on each website for the creation of the Gaza-Israel dataset. The query type column denotes whether a specified website or own query was used.

The second step was to use a precise query to get all relevant fact-checking articles from the websites. Tables 3 and 4 show the queries used for each website for both datasets. AFP Factcheck and PolitiFact provided fixed categories. For Snopes and Reuters I used queries that were similar to the provided categories, but also as precise as possible to avoid false positives. Since the Gaza-Israel dataset focuses on Gaza, similar terms such as *"Westbank"* or *"Palestine"* were not used. To distinguish between news and fact-checking articles, the additional "Fact Check" was used in the Reuters queries and only fact-checking articles were accessed on Snopes.

Website	Query	Query Type	URL
AFP Factcheck	*War in Ukraine*	Website	Category
Reuters	*Ukraine Fact Check*, *Russia Fact Check*	Own Query	Ukraine Query, Russia Query
Snopes	*Ukraine*, *Russia*	Own Query	Ukraine Query, Russia Query
Politifact	*Ukraine*, *Russia*	Website	Ukraine Category, Russia Category

Table 4: An overview of the queries used for the manual data collection on each website for the creation of the Ukraine-Russia dataset. The query type column denotes whether a specified website or own query was used.

The third step consisted of selecting all relevant articles within a query, i.e. articles that covered text and text-image claims in the time frame from July 1, 2024 to April 30, 2025. Articles which contained unrelated claims, e.g. Turkish politicians meeting Syrian representatives, or covered claims about other modalities (videos, audios) were excluded.

The fourth step was to collect all relevant data from each article, i.e.: the article URL, the claim, the date of the claim and fact-check, the image URL (if applicable), the label, the label explanation and the claim type.

The last, fifth, step was to filter out claims from June 2024 to avoid data leakage. Some of the articles from July 2024 covered claims from June 2024. The manual data collection resulted in 56 claims for the Gaza-Israel dataset and 46 claims for the Ukraine-Russia dataset.

3.1.2 API Data Collection

As with manual data collection, the first step was to select professional, reliable, and independent fact-checking websites. By using Google's FactCheck Claim Search API the scope of website was already pre-determined. To ensure a standard quality of included fact-checks in the API, the articles must meet Google's guidelines for Search [24] and Youtube [25] [17]. Additionally, all of the websites that remained after all filtering steps of the pipeline are an active and verified member of at least one of the following networks or databases: the IFCN [33], the European Fact-Checking Standards Network (EFCN) [14] or the Reporters Lab of Duke University [12].

The final set of the utilized API fact-checking websites is displayed in Chapter 3.4 for the Gaza-Israel and Ukraine-Russia dataset.

Query

For the API data collection, I used the same queries as for the manual websites which did not have specified categories: *"Gaza"*, *"Israel"*, *"Ukraine"* and *"Russia"*. Since the collected claims included videos, noisy labels and duplicates, a pre-processing pipeline was necessary.

Preprocessing Pipeline

The first step was to filter out claims with dates before July 2024 and after April 2025.

The second step was to filter out claims related to a video. To identify these claims, the text of the claim and the title of the article were searched for the keyword "video". The results were manually verified by checking the fact-checking articles. Some claims might be related to videos without mentioning the keyword. These claims were filtered out after the pre-processing pipeline in the manual step "additional data collection from article".

The third step was to filter out all claims from the websites that had already been used in the manual data collection (AFP, Reuters, Snopes, PolitiFact).

The fourth step was to filter out claims with a noisy, i.e., too long label. Some websites used entire sentences instead of a precise and short label. To be consistent with the labels from the websites of the manual data collection, all claims with labels longer than three words were removed.

In a fifth step, the claims from both queries were concatenated before filtering out duplicate claims.

To remove duplicate claims, two methods were used. First, exact duplicates with the same claim text were removed. Second, a text-embedding approach with subsequent manual verification was used to identify claims with the same content but different formulations. For all manual and API claims, text embeddings were created to identify duplicates *within* all API claims but also *between* all manual and API claims. To be consistent with using a MLLM from Google (*Gemini 2.0 Flash-Lite*) in the DEFAME system, I also applied two text-embedding models from Google [20]: *gemini-embedding-exp-03-07* and *text-embedding-004* with different similarity thresholds (0.65, 0.7, 0.75, 0.8, 0.85). The *text-embedding-004* with a threshold of 0.8 produced the best results after manually verifying each combination. Since a claim could display similarity above the threshold with multiple claims, a connected component approach was used. The idea is that if several claims form one connected component, they might be duplicates and could be represented as one claim. Even though a threshold of 0.8 produced the best results, I also manually verified 30-50 pairs of similar claims below the threshold for each dataset.

The text-embedding approach was first applied *within* all API claims, then *within* all manual claims, and then *between* all API and manual claims. First, duplicates within all API claims should be identified. Second, the text-embeddings were used as a sanity check for all manually collected claims: they should be distinct claims and no duplicates should exist. For both datasets, the sanity check was successful. Third, only API claims that are distinct from manual claims should be added to the dataset. If an API and manual claim were duplicates, the manual claim was kept since the data collection for it was already done.

Additional Manual Data Collection from Article

Since the API data collection did not contain label explanations or image URLs, an additional manual data collection step was necessary. This step also served as another manual verification step to filter out duplicates or irrelevant claims, e.g. related to videos or audios, that have not been filtered out by the previous steps. Just as in the manual data collection, I here, too, added information about the type of each claim (text-only, normal image, AI image, altered image).

3.2 Post-Processing: Label Aggregation, Claim Reformulation and Image Scraping

Label Aggregation

After concatenating the manual and API claims, the post-processing pipeline is applied. The labels are aggregated into a final set of coherent labels, the claims are reformulated to a coherent format, and images are scraped.

The labels from the fact-checking websites need to be aggregated, because for both the Gaza-Israel and Ukraine-Russia dataset the original number ($n > 10$) is too large for any prediction model. Further, it makes sense to aggregate similar original labels, such as *unproven* or *no evidence*, into an aggregated label, e.g., *Not Enough Information (NEI)*. With regard to the number of labels to aggregate, I took inspiration from the labels of existing multimodal fact-checking datasets which are displayed along the labels of my datasets in Table 5.

Dataset Name	Paper Date	Labels
MOCHEG (Multimodal Fact-Checking and Explanation Generation) [68]	07/2023	Supported, Refuted, Not Enough Information (NEI)
VERITE (VERification of Image-TExt pairs) [50]	01/2024	Truthful, Out-of-Context, MisCaptioned Images
CLAIMREVIEW2024+ [1]	12/2024	Supported, Refuted, Misleading, Not Enough Information (NEI)
Gaza-Israel Dataset		True, False, Misleading, Not Enough Information (NEI)
Ukraine-Russia Dataset		True, False, Misleading, Not Enough Information (NEI)

Table 5: An overview of the labels used in existing and my new multimodal fact-checking datasets. The table shows the datasets' names, publication date, and labels.

To guide the label aggregation, I checked whether a fact-checking methodology existed for each website and, if so, whether the labels were clearly defined and explained. Only if both conditions are met, a claim is kept in the dataset and used for label aggregation. To meaningfully aggregate the labels, it is necessary to understand how a website came to an existing label and whether websites have a shared or differing understanding of the same label.

Table 16 in Appendix C provides an overview of the fact-checking websites which have a methodology *and* label definitions. All claims from these websites were kept and used for the label aggregation for both datasets. Table 17 in Appendix C provides an overview of the fact-checking websites without label definitions or a methodology. All claims from these websites were removed from the datasets.

This last filtering step resulted in a final dataset size of n = *100* for the Gaza-Israel dataset and n = *79* for the Ukraine-Russia dataset.

The text-only claims and the claims with normal images were re-labeled to one of the four labels displayed in Table 5. All claims with AI-generated or altered images were re-labeled to the label *False*. This made a reformulation into a consistent format necessary, which is explained next.

Claim Reformulation

Most of the original claims do not specify the date and location of the event or the context they are referring to, e.g., *"Americans bombarded Yemen in reprisal after Yemen launched an attack on Israel's capital."* or *"Lebanon shot down its enemy's helicopter"*.

To ensure a consistent format of all claims and make claims as less ambiguous as possible, the following rules were applied: (1) For all claim types, if applicable, the (alleged) date and location of an event were added. For the date, either the specific day or month of an event was used. (2) For all claims with normal images, each claim starts with *"This image shows...."*. (3) For all claims with AI-generated and altered images, the adjective *"authentic"* was added. Only using the phrase *"The image shows"* would result in ambiguous claim-label relationships for these claim types. One could have also reformulate these claim types to explicitly state the AI-generated or altered/photoshopped nature of the images, but this might have simplified the verification task.

Table 6 shows an example of the original and reformulated claim for each claim type.

Image Scraping

Claim Type	Dataset	Original Claim	Reformulated Claim
Text-Only Claim	Gaza-Israel	Netanyahu lifted off out of Israel abandoning his people to Irans ballistic missiles.	Netanyahu left Israel and fled to Poland amid Iran's missile attack on Israel on October 1, 2024.
Claim with Normal Image	Gaza-Israel	Americans bombarded Yemen in reprisal after Yemen launched an attack on Israel's capital.	This image shows the U.S. bombarding Yemen on December 22, 2024 in reprisal after Yemen launched an attack on Israel's capital.
Claim with AI Image	Gaza-Israel	A PHOTO FOR THE HISTORY BOOKS MEA airline landing in Beirut International Airport as Israeli rains fire on the airport.	This is an authentic image of Israel bombing Beirut's International Airport in October 2024.
Claim with Altered Image	Ukraine-Russia	70,000 Ukrainian soldiers in the Kursk region died in vain. The UK poured hundreds of millions of pounds into Zelensky's crushing failure.	This image shows a screenshot of an authentic Hull Daily Mail's frontpage from March 13, 2025 saying '70,000 Ukrainian soldiers in the Kursk region died in vain. The UK poured hundreds of millions of pounds into Zelensky's crushing failure'.

Table 6: Examples of original and reformulated claims for each claim type along with the claims' dataset origin.

The final step was to scrape the images and store them locally.

3.3 Preliminary Experiments: Wrong Ground Truth Labels and Binary Label Aggregation

Preliminary DEFAME experiments revealed reliability problems with the aggregated ground truth labels on both datasets. Qualitative analysis on the Gaza-Israel dataset yielded an initial pattern: Claims with normal images that were out of context, were correctly predicted as *Misleading* by DEFAME, but were wrongly labeled as *False* by fact-checking websites.

To get an overview of the scope of possible wrong ground truth labels in both datasets, I systematically checked the correctness of each claims' label for both datasets. Table 18 in Appendix C gives an overview of the label definitions I based the check on which were the same that were used to prompt the MLLM in the preliminary DEFAME experiments.

The check revealed a clear pattern for both datasets: The overwhelming majority of all identified wrong labels were claims with normal images which were labeled as *False*, but should have been labeled as *Misleading*. This amounted to 16 out of all 19 incorrect claims for the Gaza-Israel dataset and five out of 7 incorrect claims for the Ukraine-Russia dataset. Further, the labels *False* and *NEI* have been used very inconsistently between different fact-checking websites on the Ukraine-Russia dataset with regards to unsubstantiated claims, e.g., *Ukraine is selling half of the armaments it receives from the United States.*

If fact-checking websites label these claims as *False* they often justify that by the absence of evidence, which is a clear indication that one could also label it as *NEI*. This pattern was visible for around 15 of the 79 claims.

Hence, for each dataset around 20 claims possibly have wrong ground truth labels, which is around 20% (Gaza-Israel) and 25% (Ukraine-Russia) of all claims.

Ideally, one would have recruited three to five human experts, i.e. professional fact checkers, with the label definitions and clear instructions to relabel both datasets. This would have validated my initial analysis and allowed to calculate inter-coder reliability metrics. Since I did not have the resources for this, I aggregated the *False*, *Misleading* and *NEI* label into one *False* label to still enable DEFAME experiments on my datasets.

All three labels can be viewed as sub-labels of a bigger *False* label: the claim is wrong either because it is demonstrably false, taken out of context or because no evidence supports it. One might argue that *NEI* is a distinct label, yet the Ukraine-Russia dataset has shown that the fact-checking websites have not consistently differentiated between the labels *False* and *NEI*.

As a result, the multi-class classification task has become a binary classification task. The reduction of noise and errors in my original four labels for both datasets is a future task. Methods to efficiently identify erroneous labels and reduce costly human work, such as verbalized confidence prompting [66, 60] and confidence learning [45, 46], are discussed in Chapter 5.

3.4 Descriptive Statistics of the Datasets

This part gives an overview of the key descriptive statistics of the Gaza-Israel and Ukraine-Russia dataset. Figure 4 displays the absolute and relative frequency of the claims collected manually and per API. The distribution is almost the same for both datasets: more than half of the claims were collected manually.

Yet, looking more closely at the number of collected claims per fact-checking website in Figure 5, we can see different patterns. The distribution of claims among websites is

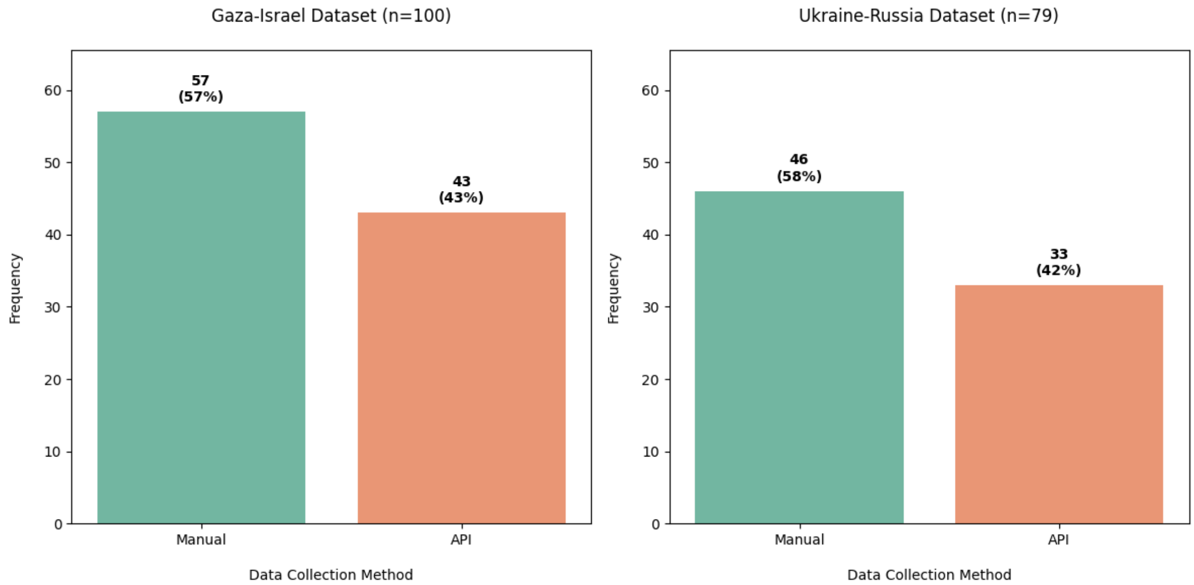

Figure 4: The absolute and relative frequency of the claims per data collection method (Manual vs. API Data Collection) for the Gaza-Israel and Ukraine-Russia dataset.

more even for the Ukraine-Russia dataset - both for the manual and API data collection. In contrast, for the Gaza-Israel dataset, more than half of all API claims (26 of 43) were collected from misbar.com and manual claims were mostly collected from Reuters and AFP Factcheck.

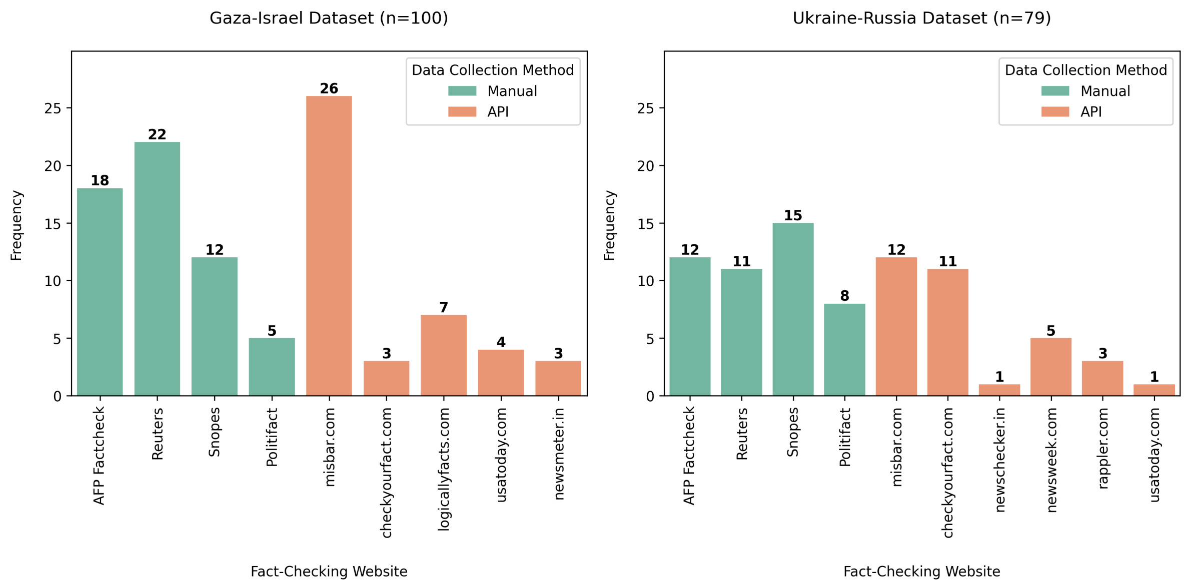

Figure 5: The number of claims per fact-checking website by data collection method (Manual vs. API Data Collection) for the Gaza-Israel and Ukraine-Russia dataset.

Figure 6 displays the claim type distribution for both datasets. The distribution is almost inverse: Whereas roughly two out of three claims in the Gaza-Israel dataset contain images, only one of three claims does for the Ukraine-Russia dataset. Almost half of all claims in the Gaza-Israel dataset relate to a normal image, but only 22% of the claims in the Ukraine-Russia dataset. The ratio of the new claim types (AI and altered images) is small among both datasets, but still noticeable: 16% and 11%. Yet, their amount is also limited by the chosen topics and time frame within the data collection.

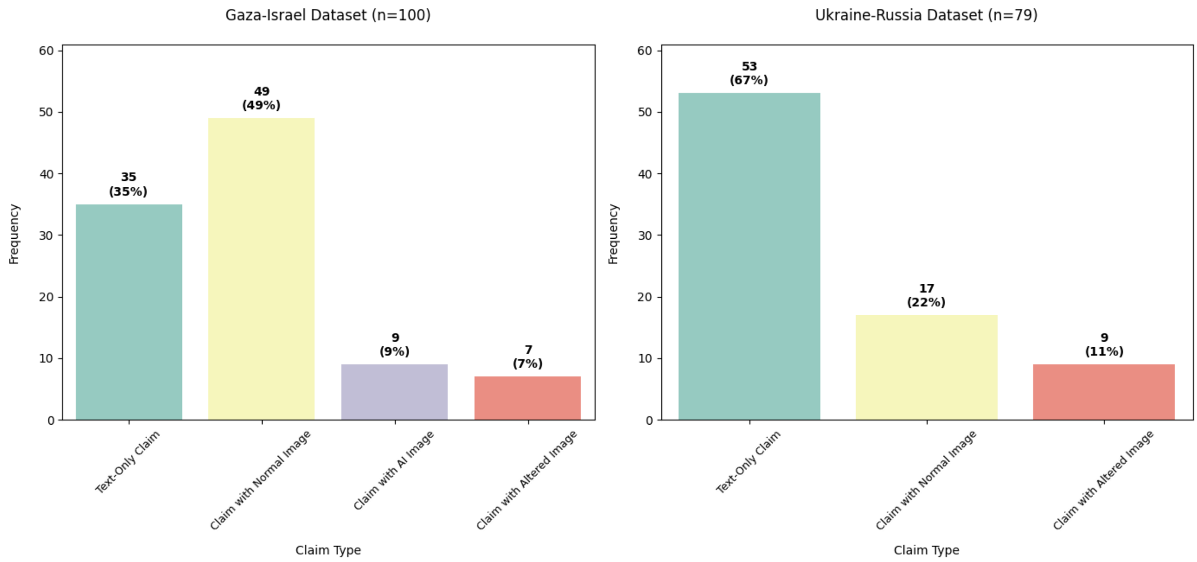

Figure 6: The absolute and relative frequency of the four claim types for the Gaza-Israel and Ukraine-Russia dataset.

Figure 7 shows the distributions of the original four and final two labels for both datasets. Even though the DEFAME experiments in Chapter 4 only use the final binary labels, the original (problematic) four labels are also visualized for transparency. Both the original and final label distributions are quite similar among both datasets. The final two labels are extremely imbalanced: only 6% and 11% of the claims are true. The most likely reason for this is the nature of viral claims on social media platforms and the selection of claims by fact-checking websites: they mostly select claims that are viral on social media, which in turn are often wrong.

Finally, Figure 8 displays the distribution of the final labels grouped by the claim type

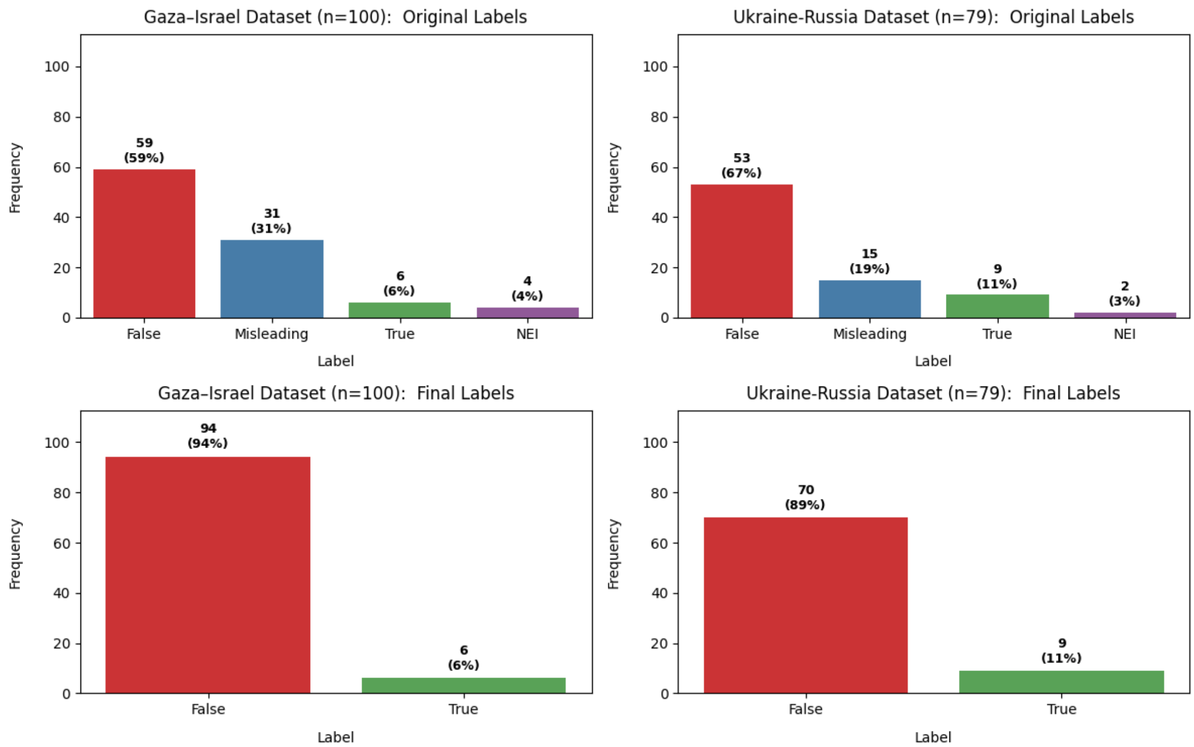

Figure 7: The absolute and relative frequency of the original four and final two labels for the Gaza-Israel and Ukraine-Russia dataset.

for both datasets. We can see a clear pattern, which is even more pronounced for the Ukraine-Russia dataset: eight out of all nine true claims are text-only claims. For the Gaza-Israel dataset four out of all six true claims are text-only claims. These distributions will become important for the claim type-level evaluations of all tested DEFAME variants reported in Chapters 4.2.1 and 4.3.1.

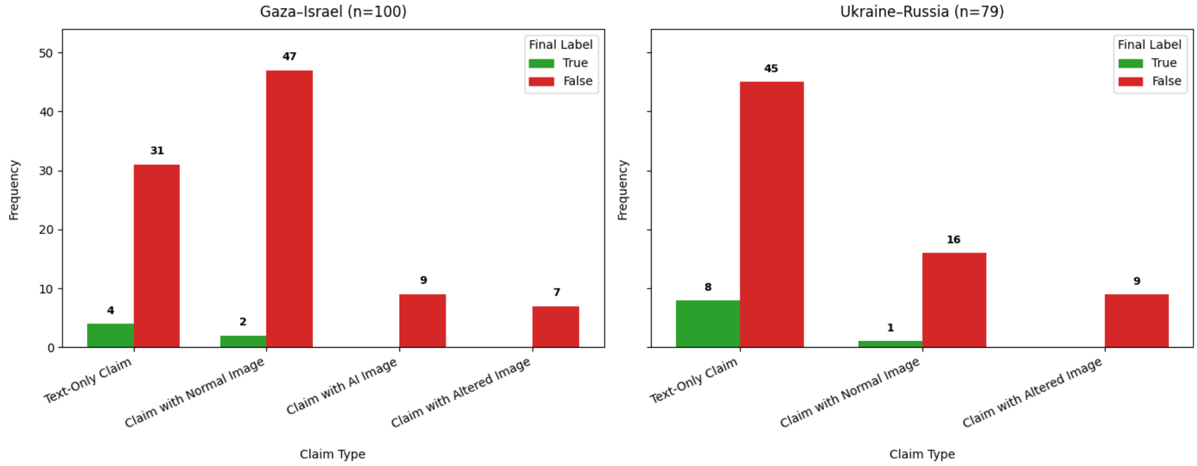

Figure 8: The absolute frequency of the final two labels grouped by the claim type for the Gaza-Israel and Ukraine-Russia dataset.

3.5 Ethical and Legal Context of the Data Collection

Salganik's collection of ethical considerations served as a guideline for data collection and handling here [55]. This has three larger dimensions: Firstly, respect for law and public interest does not apply in my case since the collected data is considered public data, therefore, their handling does not fall under the scope of the General Data Protection Regulation [15]. The second dimension concerns anonymity and confidentiality. The data collected for the purpose of this thesis are claims and/or images widely circulated by individuals or entities through social media and news outlets. The claims used here are taken from fact-checking websites, if collected manually, and from Google's FactCheck Claim Search API, if collected automatically. All claim texts are reformulated to a coherent format, so their original formulation is not identifiable anymore and not put through the DEFAME pipeline. In any case, social media users are ideally aware that their posts are publically available data. In this regard, no informed individual consent was obtained by users. The images included in the datasets depicted places or public figures, which limits privacy concerns. Taking all information like original claim text, image, posting date and platform together, there remains a remote chance that users could be reidentified. This, however, is also true for the claim verification done by the respective fact-checking websites. The research quality standard of transparency, however, outweighs further measures of anonymization or stripping of information.

4 Evaluation: DEFAME

This chapter evaluates the performance of the DEFAME system on the two new datasets. First, the experimental setup and evaluation strategy are explained in chapter 4.1. After that, the results for each dataset are presented and analyzed quantitatively and qualitatively starting with the Gaza-Israel dataset in chapter 4.2, followed by the Ukraine-Russia dataset in chapter 4.3. Finally, the limitations of my results are discussed in chapter 4.4.

4.1 Experimental Setup and Evaluation Strategy

The experimental setup consists of testing five DEFAME variants each for three replication runs on both datasets. The five variants are the same that were used in the tool ablations of the original DEFAME paper [1]: DEFAME (default), without web search, without reverse image search, without image search and without geolocate.

Originally, an MLLM without tools access (Gemini 2.0 Flash-Lite) should have served as a baseline and been compared to the DEFAME variants. This would have tested whether there was data leakage in the Gaza-Israel and Ukraine-Russia dataset. Yet, this approach assumed that no ground truth problems existed in the labels from the fact-checking websites. As elaborated in Chapter 3.3, I have aggregated the *False, Misleading* and *NEI* labels into one *False* label to deal with the problem, since a complete re-labeling of both datasets and subsequent human validation would have been beyond the scope and available resources of this thesis. Since the *NEI* label is now part of the *False* label an MLLM baseline does not make sense. Preliminary experiments showed that the MLLM is *correct for the wrong reasons*: the model does not have any evidence about the claims, since they are in the future. Due to the extreme label imbalance, the model could just predict *Not Enough Information* for every single claim and would be correct in 94 of 100 cases in the Gaza-Israel and 70 of 79 cases in the Ukraine-Russia dataset. Below are some MLLM justifications that illustrate the problem of being *correct for the wrong reasons*:

> *"The claim refers to a future date (February 2025). It is impossible to verify the accuracy of a claim about a future event."*

> *"The claim is false because the date provided (July 8, 2024) is in the future."*

> *"The claim is speculative as it refers to events in February 2025. It is impossible to verify the claim's veracity at this time because the events have not yet occurred."*

This pattern was also observed in the two MLLM baselines (GPT-4o, GPT-4o Chain-of-Thought (CoT)) evaluated on the CLAIMREVIEW2024+ dataset, which even had four labels, in the original DEFAME paper: both baselines heavily overpredicted the *NEI* label. More than half of all claims were predicted as *NEI* (GPT-4o: $n = 170$; GPT-4o CoT $n = 165$), even though only 20 of the 300 claims had the label *NEI*.

To make sure the problem of being *correct for the wrong reasons* did not occur in the DEFAME variants, a qualitative analysis of a random sample of all true negative predictions for both datasets was performed in chapters 4.2.2 and 4.3.2. The results from the qualitative analysis did not indicate any problems: the DEFAME variants successfully retrieved evidence and used it for its predictions.

The default variant has access to all four tools and dynamically selects the tools (based on the claim type) and the search depth (based on the retrieved evidence) for each claim. In the original paper, the ablation study was performed on the MOCHEG, VERITE and CLAIMREVIEW2024+ datasets, which contain a maximum of two claim types: text-only claims and claims with normal images. Further, the evaluation was only performed on a dataset-level and not on a claim-type level. In contrast, my evaluation covers two new claim types and features both levels. As a result, a more fine-grained analysis on the relationship between claim types and tool use is possible. The number of replication runs resembles the original DEFAME paper and balances the need for uncertainty estimation and computational costs. An overview of the computational costs for each DEFAME variant for one replication run is provided in Appendix D. Table 19 provides the costs on the Gaza-Israel dataset and table 20 on the Ukraine-Russia dataset. For a rigorous statistical measurement of DEFAME's reliabilty, the sample sizes of my datasets are too small and would have needed to be planned a priori based on target margin of errors and confidence levels [43]. This was not the primary goal of my thesis, but my approach yields valuable information about DEFAME's uncertainty and can be classified as a variance-based uncertainty estimation [30].

The evaluation strategy must take the label distribution of the datasets and the application context into account. The original DEFAME paper uses accuracy as the primary evaluation metric across the tested datasets [1]. Yet, due to the extreme label imbalance of both new datasets, this metric is unsuitable. Further, it does not reflect on the costs of false positive and false negative predictions if DEFAME were to be applied in real-world fact-checking, e.g., on social media platforms.

My evaluation reports metrics on two levels: for each dataset and for the claim types within each dataset. This is due to the fact that I want to compare the performance of different DEFAME variants between the datasets, but also across the claim types. Tables 21 and 22 provide an overview of the metrics, their notation, mathematical formulas, and range of possible values for the dataset and claim type level.

The choice of my evaluation metrics was based on three criteria: the effectiveness of a metric (to choose the best classifier), the ease of interpretation of a metric, and the ability of a metric to penalize false positive predictions. Simulation and empirical studies have tested the effectiveness of various evaluation metrics for binary classification on imbalanced datasets [7, 28, 54, 8]. *Matthews Correlation Coefficient (MCC)* consistently performs as an effective metric across all studies and is easy to interpret with values ranging from -1 to +1. [7] make a case for also using Cohen's Kappa and the Geometric Mean, whereas the results of [8] suggest to avoid Cohen's Kappa. The area under the Receiver Operating Characteristic (ROC) curve [28] and the Precision-Recall (PR) curve

[54] also show good performance, yet are not applicable to my case, due to the lack of predicted probabilities with LLM-based classification. Since the Geometric Mean is more difficult to interpret, *Balanced Accuracy (BA)* will be used as it accounts for label imbalance and is easy to interpret [2].

For ease of interpretation, I decided to also report absolute numbers: the number of correct (*Correct*), wrong (*Wrong*), false positive (*FP*) and false negative (*FN*) predictions. To measure the variability of errors, I also report the number of predictions that were consistently predicted wrongly across all three replication runs (*Consistently Wrong*).

Lastly, I decided to add metrics that penalize false positive predictions. If an automated fact-checking system like DEFAME would be deployed in the real-world, the costs of false positive and negative predictions become crucial. A false positive prediction would label a false claim as truthful, and contribute to the spread of misinformation, whereas a false negative prediction would label a true claim as false and contribute to suppression of free speech or censorship. I argue that in the context of wars where misinformation can be used to demonize populations and, in the worst case, legitimize war crimes, false positive predictions are very costly and should be avoided in automated systems. In addition, real-world fact-checking usually only flags false claims, but does not directly delete them which reduces the costs of false negative predictions.[1]

The specificity or *True Negative Rate (TNR)* measures the proportion of actual negative cases that the model correctly identifies. The higher the *TNR*, the better the model is at avoiding false positive predictions. Lastly, I decided to also report the *macro-averaged* F_1*-score (Macro F1)* on a dataset-level and the *precision* score for the *True* label *(Precision (True)* on the claim-type level. The F_1-score is the harmonic mean of precision and recall and is commonly used to evaluate machine learning models on imbalanced datasets. Macro-averaging this score means to treat the extremely imbalanced labels equally, which

[1] An example is the (former) cooperation between Meta and the fact-checking websites Politifact [51], where false claims are flagged after a fact-check but can still be viewed by users, e.g. https://www.facebook.com/reel/1339720690723430, https://www.facebook.com/johnsmithmarketingnashville/posts/1292593442038177/.

results in penalizing false positive predictions. Finally, *precision* measures the proportion of true positives out of all positive predictions and thus indicates the model's ability to avoid false positive predictions.

After the quantitative analysis, I employ a qualitative analysis to validate DEFAME's results and analyze its failures in depth. The steps are the same for both datasets: First, I perform a sanity check and randomly sample of true negative predictions. Since the original label *NEI* has been aggregated to *False*, it could be possible that the model *is correct for the wrong reasons*: it could not retrieve relevant evidence from the internet, and predicts *False* due to not having enough evidence. Second, I analyze those claims which were consistently predicted wrongly (*Consistently Wrong (n)*. Third, I analyze the claims with the highest error rate across all DEFAME variants and runs. These two steps could provide valuable information about the limitations of DEFAME. Fourth and last, I analyze the higher number of false negative predictions for the DEFAME variant without web search (w/o Web Search).

4.2 Gaza-Israel Dataset

4.2.1 Quantitative Analysis

Table 7 provides an overview of the dataset-level performance of the five tested DEFAME variants on the Gaza-Israel dataset. Overall, the performance does not substantially differ between the DEFAME variants except for the variant without web search. Further, the standard deviations for most metrics across all DEFAME variants are rather low indicating the robustness of the system. The highest variability of wrong predictions occurs for the variants without reverse image search and without web search. For the former, the number of false positives varies across runs, whereas for the latter both the number of false positives and negatives vary. It is quite interesting to see that removing web search results in a lower mean of false positives, which results in the highest *TNR*, but also in lower scores for *Balanced Accuracy* and *Matthews Correlation Coefficient* compared to the other variants. To better understand the higher number of false negatives for the

Metric	DEFAME (default)	w/o Web Search	w/o Reverse Image Search	w/o Image Search	w/o Geolocate
BA	0.93 ± 0.01	0.62 ± 0.07	0.90 ± 0.04	0.90 ± 0.05	0.92 ± 0.01
MCC	0.53 ± 0.01	0.17 ± 0.09	0.49 ± 0.04	0.49 ± 0.05	0.49 ± 0.03
TNR	0.87 ± 0.01	0.90 ± 0.05	0.86 ± 0.04	0.86 ± 0.02	0.84 ± 0.02
Macro F1	0.71 ± 0.01	0.58 ± 0.04	0.70 ± 0.03	0.69 ± 0.02	0.68 ± 0.02
Correct (n)	86 ± 1	86 ± 4	86 ± 3	86 ± 1	84 ± 2
Wrong (n)	12 ± 1	13 ± 4	13 ± 3	13 ± 2	15 ± 2
Consistently Wrong (n)	2	2	6	1	2
FP (n)	12 ± 1	9 ± 4	13 ± 4	13 ± 2	15 ± 2
FN (n)	0 ± 0	3 ± 3	0 ± 1	0 ± 1	0 ± 0

Table 7: Dataset-level performance metrics for the tested DEFAME variants on the Gaza-Israel Dataset ($n = 100$). Values are reported as mean ± standard deviations over the three replication runs for each DEFAME variant. The abbreviations, formulas and range of values of the metrics are explained in Table 21.

DEFAME variant without web search, these samples will be analyzed qualitatively in 4.2.2.

The number of consistently wrongly predicted claims across all three replication runs is the highest for the DEFAME variant without reverse image search with $n = 6$. One might expect that claims with images are affected more, yet the claim-type level metrics in Table 8 reveals that half of the claims are text-only claims. The other DEFAME variants have lower numbers of consistently wrongly predicted claims, yet a look on the claim-type level in Table 8 reveals an interesting pattern: For all three DEFAME variants with two consistently wrong predictions (default, w/o Web Search, w/o Geolocate), the distribution of claim types is the same: one claim with a normal image and one claim with an altered image. If these claims are the same across the different DEFAME variants, this could indicate limitations of the system. The reasons for failures and the limitations of DEFAME will be qualitatively analyzed in 4.2.2.

Moving to the performance of the DEFAME variants on a claim type level we can see in Table 8 that claims with altered images are the most difficult to predict for the default variant followed closely by claims with normal images and text-only claims. On average two out of seven claims are false positive predictions within the altered images category. The ratio of false positive predictions is similar for text-only claims (4 of 35; 11%) and

Claim Type	Metric	DEFAME (default)	w/o Web Search	w/o RIS	w/o Image Search	w/o Geolocate
Text-Only Claims (*n=35*)	BA	0.94 ± 0.02	0.53 ± 0.04	0.94 ± 0.02	0.94 ± 0.01	0.93 ± 0.03
	MCC	0.66 ± 0.06	0.04 ± 0.09	0.66 ± 0.06	0.68 ± 0.03	0.65 ± 0.08
	TNR	0.87 ± 0.03	0.89 ± 0.07	0.87 ± 0.03	0.88 ± 0.02	0.86 ± 0.05
	Precision (True)	0.50 ± 0.07	0.12 ± 0.11	0.50 ± 0.07	0.52 ± 0.04	0.49 ± 0.09
	Correct *(n)*	31 ± 1	28 ± 2	31 ± 1	31 ± 1	31 ± 2
	Wrong *(n)*	4 ± 1	7 ± 2	4 ± 1	4 ± 1	4 ± 2
	Consistently Wrong *(n)*	0	0	3	1	0
	FP *(n)*	4 ± 1	3 ± 2	4 ± 1	4 ± 1	4 ± 2
	FN *(n)*	0 ± 0	2 ± 2	0 ± 0	0 ± 0	0 ± 0
Claims w/ Normal Images (*n=48*)	BA	0.93 ± 0.01	0.80 ± 0.14	0.83 ± 0.13	0.85 ± 0.15	0.93 ± 0.01
	MCC	0.47 ± 0.04	0.39 ± 0.16	0.34 ± 0.10	0.37 ± 0.15	0.44 ± 0.02
	TNR	0.87 ± 0.02	0.92 ± 0.04	0.84 ± 0.06	0.86 ± 0.03	0.86 ± 0.01
	Precision (True)	0.25 ± 0.04	0.28 ± 0.09	0.18 ± 0.04	0.21 ± 0.08	0.23 ± 0.02
	Correct *(n)*	42 ± 1	45 ± 2	40 ± 2	42 ± 1	42 ± 1
	Wrong *(n)*	6 ± 1	4 ± 2	8 ± 3	7 ± 2	7 ± 1
	Consistently Wrong *(n)*	1	1	2	0	1
	FP *(n)*	6 ± 1	4 ± 3	8 ± 3	6 ± 2	7 ± 1
	FN *(n)*	0 ± 0	0 ± 1	0 ± 1	0 ± 1	0 ± 0
Claims w/ AI Images (*n=9*)	BA	0.96 ± 0.06	1.00 ± 0.00	0.96 ± 0.06	0.89 ± 0.01	0.80 ± 0.16
	TNR	0.96 ± 0.06	1.00 ± 0.00	0.96 ± 0.06	0.89 ± 0.01	0.80 ± 0.16
	Correct *(n)*	9 ± 1	9 ± 1	8 ± 1	8 ± 1	7 ± 2
	Wrong *(n)*	0 ± 1	0 ± 0	0 ± 1	1 ± 0	2 ± 1
	Consistently Wrong *(n)*	0	0	0	0	0
	FP *(n)*	0 ± 1	0 ± 1	0 ± 1	1 ± 0	2 ± 1
	FN *(n)*	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0
Claims w/ Altered Images (*n=7*)	BA	0.71 ± 0.15	0.67 ± 0.22	0.86 ± 0.00	0.71 ± 0.15	0.71 ± 0.15
	TNR	0.71 ± 0.15	0.67 ± 0.22	0.86 ± 0.00	0.71 ± 0.15	0.71 ± 0.15
	Correct *(n)*	5 ± 1	5 ± 2	6 ± 0	5 ± 1	5 ± 1
	Wrong *(n)*	2 ± 1	2 ± 2	1 ± 0	2 ± 1	2 ± 1
	Consistently Wrong *(n)*	1	1	1	0	1
	FP *(n)*	2 ± 1	1 ± 1	1 ± 0	2 ± 1	2 ± 1
	FN *(n)*	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0

Table 8: Claim- type level performance metrics for the tested DEFAME variants on the Gaza-Israel Dataset ($n = 100$). Values are reported as mean ± standard deviations over the three replication runs for each DEFAME variant. The abbreviations, formulas and range of values of the metrics are explained in Table 22.

claims with normal images (6 of 48; 12.5%). In contrast, the mean *precision* for the 'True' label is only half as high for claims with normal images (0.25) as it is for text-only claims (0.50). Yet, when interpreting this metric one needs to account for the low number of true claims in both categories: two among claims with normal images and four among text-only claims.

Removing the web search tool has the highest impact on text-only claims. The *preci-*

sion of the *True* label drops from 0.50 to 0.12 and the mean number of wrong predictions increases from four to seven. This is expected, since the most relevant evidence to verify text-only claims is usually retrieved from newspaper articles and the only available tool for this claim type is image search now (Reverse Image Search and Geolocate do not work with text-only claims). Yet, still 28 out of 35 claims are correctly predicted on average which indicates that the model retrieves enough relevant evidence through image search. For claims with normal images the removal of web search even slightly improves the performance. One reason might be that less conflicting or ambiguous evidence is retrieved. For claims with AI or altered images the removal of web search does not change the performance.

Removing reverse image search yields expected but also suprising findings. As expected, the removal decreases the performance on claims with normal images. It indicates that to verify these claims retrieving evidence from sources that contain the exact image is relevant. Surprisingly, the tool removal does not impact the performance on claims with AI and altered images. It indicates that the model can still retrieve enough relevant evidence with the other tools.

The last two DEFAME variants, without image search and geolocation, generally do not have an impact on the performance of any claim type. Removing the geolocation tool only slightly decreases the performance on claims with AI images, yet this change should be taken with caution due to the small sample size of this claim type.

Overall the DEFAME variants perform very well on the two new claim types introduced in the Gaza-Israel dataset: claims with AI and altered images. Even though the sample sizes are small, the performance on both claim types is very consistent across all DEFAME variants. In fact, the claims with AI images seem to be the easiest claim type to predict overall. It indicates that DEFAME is a robust system that can handle new modality types of digital misinformation.

4.2.2 Qualitative Analysis

Sanity Check: Correct Predictions for the Wrong Reasons?

The first step of the qualitative analysis was to validate the true negative predictions. Due to the aggregation of the original *NEI* label into the final *False* label, the model could be *correct for the wrong reasons*: it does not have enough information due to failed evidence retrieval and correctly predicts *False*. The total number of experiments on the Gaza-Israel dataset is 15 (5 DEFAME variants $\times$ 3 replication runs). Each experiment produces 100 fact-checking reports which are around three to five pages long each. Of these 100 reports only the reports of the true negative predictions are of interest. Most of the experiments resulted in 80 to 85 true negative predictions. As a result, around 1.200 relevant fact-checking reports were produced ($80 \times 5 \times 3$). Thus, a sampling strategy was necessary: I decided to randomly sample a total of $n = \mathit{20}$ reports. This is a quarter of the 80 true negative predictions, which most experiments revolve around. Next, I needed to decide which DEFAME variants to investigate and which replication run for each variant to use. Since the quantitative results from Table 8 found the removal of web search to have the highest impact followed by the removal of reverse image search, I decided to sample $n = \mathit{10}$ from one run of DEFAME w/o Web Search, $n = \mathit{5}$ from one run of DEFAME w/o RIS and $n = \mathit{5}$ from one run of DEFAME (default) as a baseline. Which of the three replication runs to choose, was also decided by random chance.

For each sample, I checked the fact-checking report for potential problems with evidence retrieval, compared the model justification with the ground truth justification of my dataset and documented whether the model was correct for the wrong reasons. None of the analyzed samples revealed a problem.

Error Analysis: Which Claims were the most difficult to verify?

Table 7 shows the number of consistently wrong predictions per DEFAME variant across all three replication runs. The total number across all DEFAME variants is $n = \mathit{13}$. These 13 wrong predictions are based on ten unique claims. If a DEFAME variant repeatedly fails to predict specific claims, it might indicate limitations of the system. Table 23 in Appendix D provides a descriptive overview of the ten claims with their index, claim type, the DEFAME variant(s) within which it was consistently wrongly predicted, and the overall frequency with which the claim was wrongly predicted across all variants and replication runs. Only one of the claims (index: 34) is not part of the ten most frequently mispredicted claims across all variants and replication runs, which are shown in Table 24 in Appendix D.

The original DEFAME paper identified five failure modes: **label ambiguity, missed evidence due to retrieval errors, reasoning errors, premature judgement,** and **wrong ground truth labels** [1].

To analytically distinguish between wrong predictions due to external error sources and internal tool limitations, the presentation of the results will be structured by the tested DEFAME variants.

In the DEFAME default variant two claims were consistently mispredicted: a claim with an altered image (index: 26) and a text-only claim (index: 90). The former claim is predicted wrongly across all DEFAME variants and features an altered image about J.D. Vance's social media profile. There were two failure reasons: **(1) retrieval errors** and **(2) missing verification of the altered image**. The system either did not retrieve or understand the content of the referenced sources and the actual content of the altered image was not verified and compared to the retrieved evidence. The failure reason for the second claim is **(3) claim ambiguity**.

The DEFAME variant without web search featured the claim about J.D. Vance again as well as a text-only claim. The failure reasons for the former claim are the same as before. Yet, this time the results from the reverse image searches indicate that the system either hallucinates the claims' image to be contained in the referenced sources or that no evidence was retrieved and the keyword in the claim (*authentic*) was not read for the prediction. For the second claim, the failure reason was **(4) misinformation in the sources**. The news articles featured the out-of-context image from the claim that originates from a similar context (attack on Israel on October 7, 2023) when reporting about the recent Houthi attack on Israel in July 2024.

The DEFAME variant without reverse image search had the highest number of consistently wrongly predicted claims ($n = 6$). For two claims with normal images (index: 8, 34), the failure reason was a combination where **(5) the context of the retrieved evidence makes the claim highly plausible** and *tool limitation*. Both claims contain images that are very similar to the claim text, yet out-of-context: One image is also from an Israeli bombing in the same month (October 2024), but from Lebanon instead of Iran (index: 8). The other image is also from Iran, but not from an oil refinery fire instead of an Israeli bombing (index: 34). Since reverse image search is not available, the exact origin of the image cannot be traced. For two (index: 70, 77) of the three text-only claims, the failure reasons were **retrieval errors** and **(6) conflicting evidence in the sources**. With regard to the success rate of Israel's iron dome system and an earthquake in Iran, the system either wrongly retrieved or understood the evidence or the newspaper articles contained conflicting evidence. The last text-only claim (index: 89) was wrongly predicted due to a combination where **the context of the retrieved evidence makes the claim highly plausible**, **conflicting evidence in the sources** and the difficult nature of verifying non-publicly available intelligence details. The failure reasons for the sixth claim about J.D. Vance's social media profile were the same as in other variants.

The last two DEFAME variants (without geolocation and image search) also contained the just mentioned claim about J.D. Vance and the failure reasons were the same again. The remaining claim with a normal image in the variant without geolocation was wrongly predicted due to **misinformation in the sources**. The claim's out-of-context image was presented in news reports about the new context of the claim. The remaining text-only claim has a **(7) wrong ground truth label**. The prediction was correct, but the claim misses an image and should have a different label.

An overview of the results from the qualitative analysis for all claims is provided in Table 27 in the Appendix.

DEFAME w/o Web Search: Increased Number of False Negative Predictions

Finally, the higher number of false negative predictions for the DEFAME variant without web search, as shown in Table 7, was analyzed qualitatively. Removing web search increased the average number of false negative predictions from 0 to three. The reason for this is the skewed label distribution per claim type illustrated in Figure 8. Only six of the 100 claims from the Gaza-Israel dataset have the label *True*. Four of these six claims are text-only claims. The qualitative analysis confirms the importance of the web search tool to verify text-only claims. By removing the tool, DEFAME can only use image search for text-only claims and could not retrieve enough relevant evidence to verify the dates or locations mentioned in a claim. Hence, the true claims were predicted as *False*.

4.3 Ukraine-Russia Dataset

4.3.1 Quantitative Analysis

Table 9 provides an overview of the dataset-level performance for the five tested DEFAME variants on the Ukraine-Russia dataset. Overall, a similar pattern to that in the Gaza-Israel dataset emerges: the performance does not substantially differ between the DEFAME variants except for the variant without web search. Again, the standard deviations for most metrics across all DEFAME variants are rather low; confirming the robustness of the system. The variability of correct and wrong predictions is more evenly distributed among the DEFAME variants than in the Gaza-Israel dataset.

The pattern for the DEFAME variant without web search is the same as in the Gaza-Israel dataset: the number of false positive predictions slightly decreases and the number of false negative predictions strongly increases. As a result, this variant has the highest *True Negative Rate*, but lower scores for *Balanced Accuracy* and *Matthews Correlation Coefficient* again. Yet, the number of false negative predictions is more stable on this dataset with a standard deviation of 0. Just like for the Gaza-Israel dataset, the higher

Metric	DEFAME (default)	w/o Web Search	w/o Reverse Image Search	w/o Image Search	w/o Geolocate
BA	0.86 ± 0.04	0.63 ± 0.01	0.87 ± 0.08	0.85 ± 0.04	0.91 ± 0.03
MCC	0.60 ± 0.07	0.28 ± 0.03	0.59 ± 0.10	0.56 ± 0.05	0.67 ± 0.04
TNR	0.90 ± 0.01	0.93 ± 0.02	0.89 ± 0.02	0.88 ± 0.02	0.90 ± 0.01
Macro F1	0.79 ± 0.04	0.64 ± 0.02	0.78 ± 0.03	0.77 ± 0.03	0.82 ± 0.02
Correct (n)	70 ± 1	68 ± 1	70 ± 2	68 ± 2	72 ± 1
Wrong (n)	8 ± 2	11 ± 1	9 ± 1	10 ± 1	7 ± 1
Consistently Wrong (n)	0	5	3	1	3
FP (n)	7 ± 1	5 ± 1	8 ± 1	8 ± 1	7 ± 1
FN (n)	2 ± 1	6 ± 0	1 ± 2	2 ± 1	1 ± 1

Table 9: Dataset-level performance metrics for the tested DEFAME variants on the Ukraine-Russia Dataset ($n = 79$). Values are reported as mean ± standard deviations over the three replication runs for each DEFAME variant. The abbreviations, formulas and range of values of the metrics are explained in Table 21.

number of false negatives will be analyzed qualitatively in 4.3.2.

With regards to the number of consistently wrong predictions a different pattern is visible: the number is the highest for the variant without web search ($n = 5$) followed by the variants without reverse image search ($n = 3$) and geolocate ($n = 3$). Looking at the claim-type level performance in Table 10 we can see that for the variant without web search all claims are text-only claims, which is plausible and expected. With regards to the two other variants, the same pattern as in the Gaza-Israel dataset exists: the distribution of the claim types is the same. There are two text-only claims and one claim with an altered image each. Theoretically one might have expected more claims with normal or altered images among them when removing the tools reverse image search and geolocation. Again, the consistently wrong claims could indicate limitations of the system and will be analyzed qualitatively in 4.3.2.

Moving to the claim-type level performance in Table 10, we can see a different pattern for the DEFAME default variant for both datasets. In the Gaza-Israel dataset text-only claims and claims with normal and altered images were similarly difficult to predict. In the Ukraine-Russia dataset the metrics *Balanced Accuracy* and *Matthews Correlation Coefficient* indicate that text-only claims are the most difficult to predict. The difference to the other claim types is not very great, but noticeable.

Claim Type	Metric	DEFAME (default)	w/o Web Search	w/o RIS	w/o Image Search	w/o Geolocate
Text-Only Claims ***(n=53)***	BA	0.85 ± 0.04	0.58 ± 0.00	0.86 ± 0.08	0.83 ± 0.03	0.90 ± 0.03
	MCC	0.62 ± 0.07	0.18 ± 0.00	0.61 ± 0.09	0.56 ± 0.05	0.68 ± 0.05
	TNR	0.90 ± 0.01	0.91 ± 0.00	0.88 ± 0.04	0.87 ± 0.00	0.89 ± 0.00
	Precision (True)	0.60 ± 0.05	0.33 ± 0.00	0.56 ± 0.03	0.51 ± 0.02	0.59 ± 0.02
	Correct *(n)*	47 ± 1	43 ± 1	46 ± 1	45 ± 1	47 ± 1
	Wrong *(n)*	6 ± 1	10 ± 0	7 ± 1	8 ± 1	6 ± 1
	Consistently Wrong *(n)*	0	5	2	1	2
	FP *(n)*	4 ± 1	4 ± 0	5 ± 2	6 ± 0	5 ± 0
	FN *(n)*	2 ± 1	6 ± 0	1 ± 2	2 ± 1	1 ± 1
Claims w/ Normal Images ***(n=17)***	BA	0.96 ± 0.05	1.00 ± 0.00	0.96 ± 0.02	0.99 ± 0.02	0.98 ± 0.02
	MCC	0.71 ± 0.28	1.00 ± 0.00	0.63 ± 0.08	0.89 ± 0.18	0.79 ± 0.18
	TNR	0.91 ± 0.10	1.00 ± 0.00	0.92 ± 0.04	0.98 ± 0.04	0.96 ± 0.03
	Precision (True)	0.58 ± 0.38	1.00 ± 0.00	0.44 ± 0.10	0.83 ± 0.29	0.67 ± 0.29
	Correct *(n)*	15 ± 2	17 ± 0	15 ± 1	16 ± 1	16 ± 1
	Wrong *(n)*	1 ± 2	0 ± 0	1 ± 1	0 ± 1	1 ± 1
	Consistently Wrong *(n)*	0	0	0	0	0
	FP *(n)*	1 ± 2	0 ± 0	1 ± 1	0 ± 1	1 ± 1
	FN *(n)*	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0
Claims w/ Altered Images ***(n=9)***	BA	0.89 ± 0.00	0.89 ± 0.11	0.89 ± 0.00	0.82 ± 0.06	0.89 ± 0.00
	TNR	0.89 ± 0.00	0.89 ± 0.11	0.89 ± 0.00	0.82 ± 0.06	0.89 ± 0.00
	Correct *(n)*	8 ± 0	8 ± 1	8 ± 0	7 ± 1	8 ± 0
	Wrong *(n)*	1 ± 0	1 ± 1	1 ± 0	2 ± 1	1 ± 0
	Consistently Wrong *(n)*	0	0	1	0	1
	FP *(n)*	1 ± 0	1 ± 1	1 ± 0	2 ± 1	1 ± 0
	FN *(n)*	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0

Table 10: Claim-type level performance metrics for the tested DEFAME variants on the Ukraine-Russia Dataset (n = 79). Values are reported as mean ± standard deviations over the three replication runs for each DEFAME variant. The abbreviations, formulas and range of values of the metrics are explained in Table 22.

Removing web search as a tool yields the same patterns as in the Gaza-Israel dataset: the performance on text-only claims drops, it slightly increases for claims with normal images and stays the same for claims with altered images. And also among text-only claims the pattern is the same: the number of false positive predictions stays the same (n = *4*), but the number of false negative predictions increases from two to six compared to the default variant.

Removing reverse image search has no impact on claims with normal or altered images. This finding is different from the Gaza-Israel dataset for claims with normal images, but the same for claims with altered images. It indicates that DEFAME is able to retrieve enough relevant evidence through the remaining tools to verify multimodal claims.

The results for the last two DEFAME variants, without image search and geolocation, are the same as for the Gaza-Israel dataset: the removal of these tools generally does not have an impact on the performance of any claim type.

All DEFAME variants perform consistently well on the claims with altered images. It confirms the findings from the Gaza-Israel dataset and indicates that DEFAME is a robust system to handle new modality types of digital misinformation. In this light, it did not seem necessary to test a Manipulation Detection Tool, which could be added to the DEFAME architecture.

4.3.2 Qualitative Analysis

Sanity Check: Correct Predictions for the Wrong Reasons?

The first step of the qualitative analysis was the same as for the Gaza-Israel dataset: validating the true negative predictions. I applied the same sampling strategy before: randomly sampling a total of $n = 20$ reports. Due to its smaller size, most of the DEFAME experiments resulted in 60 to 65 true negative predictions for the Ukraine-Russia dataset. Thus, the ratio of validated reports is even higher. The ratio of the sampled reports across the DEFAME variants is the same: $n = 10$ from one run of DEFAME w/o Web Search, $n = 5$ from one run of DEFAME w/o RIS and $n = 5$ from one run of DEFAME (default) as a baseline. Which of the three replication runs to choose, was again decided by random chance. The qualitative analysis yielded no problems for any of the sampled claims.

Error Analysis: Which Claims were the Most Difficult to Verify?

Table 9 shows the number of consistently wrong predictions per DEFAME variant across all three replication runs. The total number across all DEFAME variants is 12. These 12 wrong predictions are based on seven unique claims. Table 25 in Appendix D provides a descriptive overview of the seven claims with their index, claim type, the DEFAME variant(s) within it was consistently wrongly predicted and the overall frequency the claim was wrongly predicted across all variants and replication runs. Only one of the claims (index: 25) is not part of the ten most frequently mispredicted claims across all variants and replication runs, which are shown in Appendix D in Table 26.

To analytically distinguish between wrong predictions due to external error sources and internal tool limitations, the presentation of the results will be structured by the tested DEFAME variants.

The identified failure reasons in the Gaza-Israel dataset were **(1) retrieval errors, (2) missing verification of the altered image, (3) claim ambiguity, (4) misinformation in the sources, (5) the context of the retrieved evidence makes the claim highly plausible, (6) conflicting evidence in the sources and (7) wrong ground truth label**.

In contrast to the Gaza-Israel dataset, the DEFAME default variant did not have any consistently wrong predicted claim. It indicates that the dynamic selection of tools and search depth work very well on the Ukraine-Russia dataset.

For the remaining DEFAME variants, removing web search resulted in the highest number of consistently wrongly predicted claims ($n = 5$). All claims are text-only claims. This is not surprising, since the descriptive statistics in Chapter 3.4 show that two out of three claims in the Ukraine-Russia dataset are text-only claims. For two (index: 32, 25) of the five claims, the failure reason was the tool limitation: DEFAME could not retrieve enough relevant evidence without web search. For two other claims the failure reason was external: **misinformation in the sources**. The news sources contained factually wrong, misleading and/or ambiguous headlines and content (index: 31, 18). Note that for text-only claims DEFAME is also able to access the content of news articles through image search. Yet, the results indicate that with web search in the DEFAME default variant the system could retrieve higher quality evidence. The failure reason for the last claim was **claim ambiguity** (index: 28).

The results for the DEFAME variants without reverse image search and geolocation are presented together, since they feature the same three claims (index: 31, 63, 48). Two claims are text-only claims and one is a claim with an altered image. The text-only claims seem counter-intuitive for removing these tools, yet looking into the other DEFAME variants (including DEFAME default) revealed that these claims were also wrongly predicted in two of three replication runs in all other variants. This corroborates the findings from the two investigated DEFAME variants: the failure reason was external: **misinformation in the sources**, i.e. factually wrong, and/or misleading news headlines and content. The failure reason for the remaining claim with an altered image was the **missing verification of the altered image** (index: 48).

Lastly, the DEFAME variant without image search had only one consistently wrongly predicted claim. It is the claim with the highest overall error frequency across all DEFAME variants and replication runs and the failure error is the same as in the other variants: **misinformation in the sources**, i.e. factually wrong and misleading information in the news sources. An overview of the results from the qualitative analysis for all claims is provided in Table 28 in the Appendix.

In sum, the qualitative analysis of the Ukraine-Russia dataset did not reveal new failure reasons. Table 11 compares the failure reasons identified in the original DEFAME paper [1] and the qualitative analysis of my datasets. My qualitative analysis revealed the following new failure reasons: **Missing verification of altered images, claim ambiguity, misinformation in the sources, the context of the retrieved evidence making the claim highly plausible and conflicting evidence in the sources**. Only one of the reasons (claim ambiguity) is an internal dataset problem. The other reasons are external and might be reduced by incorporating new tools in the DEFAME architecture.

DEFAME w/o Web Search: Increased Number of False Negative Predictions

Finally, the higher number of false negative predictions for the DEFAME variant without web search, as shown in Table 9, was analyzed qualitatively. Removing web search increased the average number of false negative predictions from two to six. The reason for this is the skewed label distribution per claim type illustrated in Figure 8. Only nine of the 79 claims from the Ukraine-Russia dataset have the label *True*. Eight of

Failure Reason	DEFAME Paper	Own Datasets
Wrong ground truth label	✓	✓
Retrieval errors	✓	✓
Label ambiguity	✓	×
Reasoning erros	✓	×
Premature judgement	✓	×
Missing verification of altered images	×	✓
Claim ambiguity	×	✓
Misinformation in the sources	×	✓
The context of the retrieved evidence makes the claim highly plausible	×	✓
Conflicting evidence in the sources	×	✓

Table 11: An overview of the identified failure reasons in the original DEFAME paper [1] and the qualitative analysis of my datasets, i.e. the Gaza-Israel and Ukraine-Russia. The ✓symbol means that the failure reason was existent. The × symbol means that the failure reason was non-existent.

these nine claims are text-only claims. The qualitative analysis confirms the importance of the web search tool to verify text-only claims. The pattern is the same as for the Gaza-Israel dataset: Without web search, DEFAME could not retrieve enough relevant evidence through image search to verify specific dates, locations or quotes mentioned in a claim. Hence, the true claims were predicted as *False*.

4.4 Discussion and Limitations

This part sums up the previous results by answering the research questions posed in Chapter 1 and addresses the limitations of the results.

The research questions are:

1. *How does DEFAME perform on the new Gaza-Israel and Ukraine-Russia dataset? (RQ 1)*

2. *How does DEFAME perform across the four claim types, i.e. text-only claims, claims*

with normal images, AI images and altered images? (RQ 2)

3. *How does DEFAME's performance vary across the four claim types when individual tools (web search, reverse image search, image search, and geolocation) are removed? (RQ 3)*

4. *What are DEFAME's failure reasons on the new datasets? (RQ 4)*

To answer RQs 1 and 2 the results from DEFAME's default variant is used, whereas for RQ 3 and 4 the results from all DEFAME variants is used.

With regards to RQ 1, the results indicate a good performance of DEFAME: on average 86 out of 100 claims (from the Gaza-Israel dataset) and 70 out of 79 claims (from the Ukraine-Russia dataset) are correctly predicted. DEFAME's performance is robust on both datasets as indicated by low standard deviations for most metrics. The main difference between the datasets is the higher ratio of false positives among wrong predictions on the Gaza-Israel dataset (100%). In a real-world application this would result in spreading misinformation, which could become very costly in the context of high-stake wars.

With regards to RQ 2, the results are different for the datasets. For the Gaza-Israel dataset, claims with altered images are the most difficult to verify, followed closely by claims with normal images and text-only claims. Claims with AI images are the easiest to predict. For the Ukraine-Russia dataset, text-only claims are the most difficult to predict. The difference from claims with normal and altered images is not huge, yet noticeable. Overall, the DEFAME default version (and all other versions) performs well on the two new claim types (AI and altered images) for both datasets. Even though the sample sizes are small and the results should be interpreted with caution, they indicate that DEFAME is a robust system that can handle new modality types of digital misinformation. Yet, most of the claims with AI and altered images featured public figures like Benjamin Netanyahu, Donald Trump or Greta Thunberg where newspaper articles and other fact-checks about the claims were available. As a result, relevant evidence was mostly retrieved through web search and not by verifying the actual image. For claims about less-known people or events the DEFAME might perform worse and adding a Manipulation Detection Tool might be beneficial.

With regard to RQ 3, the results are very similar for the datasets. Out of all tool ablations, removing web search makes the biggest difference and reduces the performance on text-only claims the most. For the Gaza-Israel dataset, the removal of reverse image search moderately decreases DEFAME's performance on claims with normal images, but did not impact claims with AI and altered images. For the Ukraine-Russia dataset, the removal of reverse image search did not have an effect. Similarly, the removal of image search and geolocation did not impact DEFAME's performance on any claim type for any of the datasets.

With regard to RQ 4, the qualitative analysis confirmed the failure mode of retrieval errors from the original DEFAME paper [1]. It revealed the following new failure reasons: missing verification of altered images, claim ambiguity, misinformation and conflicting evidence in sources and the context of the retrieved evidence making the claim highly plausible. These results suggest that adding new tools to the modular DEFAME architecture, such as a news credibility checker or a reflection mechanism about conflicting evidence, might improve performance. A news credibility checker could be implemented with third party ratings [72]. But first one would need to check whether the news sources with misinformation are also rated less credible by third parties than those without misinformation. Since the overall number of wrong predictions is small, the additions might only yield marginal improvement.

The main limitation of the thesis comes from the binary label aggregation due to the erroneous ground truth labels from the fact-checking websites. The binary classification task is easier than the originally planned multi-class classification with four labels. As a consequence, the presented results are likely over-optimistic.

Future work consists of re-labeling the datasets. To reduce the amount of costly human labeling, the patterns identified by my check, verbalized confidence prompting [66, 60] and

LLMs could be employed - followed by human validation.

As described in Chapter 3.3 my initial check of both datasets' labels revealed that mostly claims with normal images were mislabeled in both datasets and the original labels *False* and *NEI* are not consistently applied in the Ukraine-Russia dataset. A random or full sample of these claims could be validated by other human labelers or by prompting an LLM to label the claims. Yet, the LLM would need tool access since web, image and reverse image search is necessary to label the claims.

Verbalized confidence prompting is a method to estimate the uncertainty of LLM output using language expression [30]. In contrast to probabilistic machine learning models, such as logistic regression or naive bayes, which give out predicted probabilities, uncertainty estimation for black-box LLMs, especially closed-source LLMs, is more difficult. Verbalized confidence prompting prompts an LLM to provide its k best guesses for a prediction and the probability that each is correct (0% to 100%). This approach could be implemented within the DEFAME system by using the original four labels and their definitions in the preliminary experiments, as shown in 18. Only one prompt template would need adjustment (judge.md).[2] The result would be that the fact-checking report for each claim not only features the predicted labels, but also a confidence score for each of the four labels. These scores could be used for a sampling strategy for human review: (1) claims with high confidence disagreement, where DEFAME predicts with high confidence a different label than the original label; (2) claims with normal images where DEFAME predicts *Misleading* and the original label is *False* (one of the identified pattern described above); (3) claims where DEFAME has low confidence and is unsure between *False* and *NEI* (the other identified pattern described above); (4) a random baseline sample.

Even though this approach still requires human review and a human validation of the final, corrected labels, it is an efficient alternative to re-label both entire datasets manually and thus reduces the amount of human work.

[2] I tested out the verbalized confidence prompting method within the DEFAME code and it worked.

5 Conclusion

The goal of this thesis was to evaluate the LLM-based agentic system DEFAME on two new datasets. The datasets contain claims that have new modalities of misinformation (AI-generated and altered images), are sourced *after* the knowledge cutoff of the used MLLM (Gemini 2.0 Flash-Lite) in order to circumvent data leakage, and focus on real-world high stake contexts, where misinformation can become extremely costly: the war in Ukraine and genocide in Gaza.

First, the new datasets and DEFAME were situated within the literature in Chapter 2: the overall process of automated multimodal fact-checking, existing datasets and their problems, as well as different approaches for MLLM-based fact-checking.

Second, the data pipeline and its step to create the Gaza-Israel and Ukraine-Russia datasets were described in Chapter 3. The problem of wrong ground truth labels from fact-checking websites was revealed through preliminary experiments and addressed through binary label aggregation. Finally, descriptive statistics of both datasets were presented.

Third, DEFAME was evaluated on the new datasets. The quantitative analysis showed a very good and robust performance across all four claim types on both datasets. Whereas claims with altered images were the most difficult to verify on the Gaza-Israel dataset, it was text-only claims on the Ukraine-Russia dataset. The ablation study revealed that removing the web search tool was associated with the highest drop in performance on both datasets, primarily for text-only claims. The removal of reverse images search was associated with a low drop in performance on claims with normal images on the Gaza-Israel dataset, but did not reveal any changes on claims with AI-generated or altered images. The removal of image search and geolocation did not change DEFAME's performance on both datasets. The qualitative analysis yielded new failure reasons that were not identified in the original DEFAME paper [1]: *(1) missing verification of altered images, (2) misinformation in news sources, (3) conflicting evidence in news sources, (4)* and *the context of the retrieved evidence making the claim highly plausible*. These results suggest that adding new tools to the modular DEFAME architecture, such as a news credibility checker or a reflection mechanism about conflicting evidence, might improve performance.

Future work on the Gaza-Israel and Ukraine-Russia datasets should include identifying and removing erroneous ground truth labels. To make the re-labeling efficient and reduce manual work, the method of verbalized confidence prompting was discussed in Chapter 4.4. Once this is done, DEFAME could be compared to other agentic systems, e.g. recent MLLM-based multi-agent systems with planner and deep research agents [37].

Future work on DEFAME includes applying it to other modalities: video and audio data. Especially out-of-context, altered, deepfake and AI-generated videos contribute to the spread of misinformation in the Ukraine and Gaza wars [61, 53] and likely become even more relevant in future wars. With the video and audio understanding capabilities of SOTA MLLMs, e.g. Google's Gemini models [19], claims related to these modalities can be verified with the DEFAME pipeline.

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Usage of AI

Anthropic's Claude AI, Google's Vertex AI and OpenAI's ChatGPT were used for specific purposes in this thesis: They were used to answer questions about Python code and LaTex code to format tables in Overleaf. AI was not used for other purposes in this thesis.

Appendices

A API Keys and GitHub Repository

Table 12 and 13 provide an overview of the APIs used to create the Gaza-Isreal and Ukraine-Russia dataset and to evaluate DEFAME on both datasets. The tables list the APIs, the URLs to create API keys, and the overall costs for each API.

Task	API	URL to create API key	Implemented within DEFAME codebase	Costs
Claim scraping	Google FactCheck Claim Search API	`https://console.cloud.google.com/apis/`	✓	0€
Filter out duplicate claims with text embedding models	Gemini API	`https://console.cloud.google.com/apis/`	×	$<$ 1€

Table 12: An overview of the APIs used to create the Gaza-Israel and Ukraine-Russia dataset. The table provides the task, the API, the URL to create an API key, whether the the task was implemented within the DEFAME codebase and the API costs for the task.

DEFAME Component	API	URL to create API key	Costs
Web Search	Serper API	`https://serper.dev`	0€
Image Search	Serper API	`https://serper.dev`	0€
Reverse Image Search	Google Cloud Vision API	`https://console.cloud.google.com/marketplace/product/google/vision.googleapis.com`	0€
MLLM Gemini 2.0 Flash-Lite	Gemini API	`https://console.cloud.google.com/apis/`	10-15€

Table 13: An overview of the APIs used to run DEFAME. The table provides the DEFAME component, the API, the URL to create an API key and the total costs for the evaluation of all five DEFAME variants with three replication runs on both datasets (Gaza-Israel and Ukraine-Russia). The computational costs for a single DEFAME run for each variant is provided in Table 19 for the Gaza-Israel dataset and in Table 20 for the Ukraine-Russia dataset.

Please read DEFAME's official Github Repository (`https://github.com/multimodal-ai-lab/DEFAME`) for the specific instructions to execute DEFAME through Docker and to set up and create the necessary API keys for the DEFAME components listed in Table 13. Please note that my local folder structure and my GitHub repository (`https://github.com/FabiLochner/Master-Thesis-Automated-Multimodal-Fact-Checking`), consist of three folders (DEFAME, evaluation, gaza_ukraine_datasets). To execute DEFAME, the DEFAME folder must be accessed, API keys provided in the *config* folder and Docker must be run from the DEFAME folder. The DEFAME folder already contains both datasets in the *data* folder.

I'm trying to submit a zipped version of the three folders, yet due to the high amount of fact-checking reports and scraped media the size of the zip file (15 GB) might be to large to to upload on KUNet. In this case, please access the folders through the URL of the GitHub Repository.

B DEFAME: Excluded Fact-Checking Websites/Domains and Fact-Checking Report Example

B.1 Excluded Fact-Checking Websites and Domains

Excluded Fact-Checking URLs
snopes.com
politifact.com
factcheck.org
truthorfiction.com
fullfact.org
leadstories.com
hoax-slayer.net
checkyourfact.com
reuters.com/fact-check
reuters.com/article/fact-check
apnews.com/APFactCheck
factcheck.afp.com
poynter.org
factcheck.ge
vishvasnews.com
boomlive.in
altnews.in
thequint.com/news/webqoof
factcheck.kz
misbar.com
logicallyfacts.com
newsmeter.in/fact-check
usatoday.com/story/news/factcheck
newschecker.in/fact-check
newsweek.com/fact-check
rappler.com/newsbreak/fact-check
yahoo.com/news/fact-check

Table 14: List of excluded URLs from fact-checking websites to maintain a meaningful fact-checking setting.

Unsupported Domains
facebook.com
twitter.com
x.com
instagram.com
youtube.com
tiktok.com
reddit.com
ebay.com
microsoft.com
researchhub.com
pinterest.com
irs.gov

Table 15: List of excluded domains due to bot traffic restrictions.

B.2 Example of a Fact-Checking Report

Claim

Claim: "

This is an authentic image of Greta Thunberg from November 2024 holding a sign reading, 'Fuck Israel'."

Actions

```
reverse_search(![image 29](media/45.jpg))
web_search("Greta Thunberg sign Fuck Israel")
```

Evidence

Evidence from `reverse_search`

The reverse image search found the image of Greta Thunberg holding a sign that reads "Fuck Israel" on multiple websites. NDTV.com (https://www.ndtv.com/world-news/victims-of-hamas-massacre-could-have-been-your-friends-israel-on-greta-thunbergs-support-for-gaza-4502156) and Hindustan Times (https://www.hindustantimes.com/world-news/climate-activist-greta-thunberg-voices-support-for-gaza-israel-responds-101697949844615.html) both published articles discussing Thunberg's support for Gaza, with the image included. Ynetnews (https://www.ynetnews.com/environment/article/s1n8qjyt1x) published an article discussing Thunberg's shift from climate activism to anti-Israel activism.

Evidence from `web_search`

The New York Post (https://nypost.com/2024/12/09/world-news/greta-thunberg-bursts-into-laughter-as-she-declares-f-k-israel-at-rally/) and Yahoo News (https://www.yahoo.com/news/greta-thunberg-giggles-while-saying-095000750.html) report that Greta Thunberg declared "f–k Germany" and "f–k Israel" at an anti-Israel rally in Germany. The Telegraph (https://www.telegraph.co.uk/world-news/2024/12/10/watch-greta-thunberg-declares-f-israel-german-demonstration/) also reported on the incident.

The New York Post and Yahoo News included images of Thunberg at the rally, including one where she is laughing (

). The Telegraph also included a photo of Thunberg at the event (image:k).

Elaboration

The evidence confirms that Greta Thunberg did express the phrase "Fuck Israel" at a rally. The reverse image search confirms the existence of the image in the claim. The web search results provide additional context, indicating that the incident occurred at an anti-Israel rally in Germany. The New York Post, Yahoo News, and The Telegraph all reported on the event.

The claim is partially accurate. While the image is authentic, the claim does not specify the context of the image. The evidence suggests that the image is related to an event in Germany, not necessarily from November 2024.

The claim is partially verified.

Final Judgement

The evidence confirms that Greta Thunberg did express the phrase "Fuck Israel" at a rally. The reverse image search confirms the existence of the image in the claim. The web search results provide additional context, indicating that the incident occurred at an anti-Israel rally in Germany. The claim is partially accurate, as the image is authentic, but the context of the image is not fully specified.

`true`

Verdict: TRUE

Justification

The image of Greta Thunberg holding a sign that reads "Fuck Israel" is authentic, as confirmed by reverse image searches and reports from news outlets like NDTV.com and Hindustan Times. Furthermore, Thunberg was reported to have said "f–k Israel" at a rally in Germany, as reported by The New York Post, Yahoo News, and The Telegraph.

C Own Multimodal Fact-Checking Datasets Creation: Variables, Fact-Checking Websites and Claim Reformulation

C.1 Dataset Columns Description

This section provides detailed information on each column in the new Israel-Gaza and Ukraine-Russia datasets. The data consists of claims, fact-checking sources, contextual explanations, image metadata, and classification labels.

id

Description: Unique identifier assigned to each claim record.

Type: Integer

Example: 45

Website

Description: Source or fact-checking website from which the claim was collected.

Type: String

Example: "AFP Factcheck"'

Article_URL

Description: Direct link to the fact-checking article addressing the claim.

Type: String

Example: `https://factcheck.afp.com/doc.afp.com.372Y6CV`

Headline

Description: Title or headline of the fact-checking article.

Type: String

Example: “Lebanon explosion falsely shared online as Israeli airstrikes in Iran”

Claim_Date

Description: Date when the claim was first recorded or surfaced.

Type: String

Example: 2025-03-15

Review_Date

Description: Date when the claim was fact-checked or reviewed.

Type: String

Example: 2025-03-19

Query/Keyword

Description: Search query or keyword associated with the claim collection.

Type: String

Example: “War in Ukraine”

Original_Claim_Website

Description: Complete text in which claim appeared.

Type: String

Example: “BREAKING: Zelensky blocks access to Truth Social”

Original_Claim_Only

Description: The extracted version of the claim without additional text.

Type: String

Example: “70,000 Ukrainian soldiers in the Kursk region have been killed”

Claim

Description: Manually reformulated claim.

Type: String

Example: “This is an authentic image of Bono and Bob Geldof holding the Israeli flag outside the Israeli embassy in Dublin.“

Image_URL

Description: URL to the image associated with the claim (if any).

Type: String

Example: `https://img9.irna.ir/d/r2/2024/10/01/4/171423049.jpg?ts=1727742501971`

Image_Path

Description: Local storage path for the associated claim image.

Type: String

Example: images/ukraine_russia/3.jpg

Label_Website

Description: The label provided by the fact-checking website.

Type: String

Examples: “altered”, “false”, “true”, “satire”

Label

Description: Aggregated fact-check label for the claim.

Type: Categorical (String)

Values: *True*, *False*, *Misleading*, *NEI*

Label_Binary

Description: Binary representation of the claim's label.

Type: Boolean

Values: *True*, *False*

Context/Label_Explanation

Description: Contextual explanation or justification for the label assigned by the fact-checking website.

Type: String

Example: "Reverse image search confirms the image is from 2014, unrelated to the claim."

Text_Only_Claim

Description: Indicates whether the claim is purely textual (without images).

Type: Boolean

Values: True or False

Normal_Image

Description: Indicates whether the associated image is a normal, unaltered photo.

Type: Boolean

Values: True or False

AI_Generated_Image

Description: Indicates whether the claim's image is AI-generated.

Type: Boolean

Values: True or False

Altered_Image

Description: Indicates whether the claim's image has been digitally manipulated or altered.

Type: Boolean

Values: True or False

Data_Collection_Type

Description: Method used for gathering the data (e.g., manual collection, automated scraping).

Type: String

Values: "Manual", "API"

C.2 Used and Excluded Fact-Checking Websites for Both Datasets

Website	Data Collection Type	Dataset	Methodology URL
AFP Factcheck	Manual	Gaza-Israel, Ukraine-Russia	factcheck.afp.com/stylebook
Reuters	Manual	Gaza-Israel, Ukraine-Russia	reuters.com/fact-check/about
Snopes	Manual	Gaza-Israel, Ukraine-Russia	snopes.com/fact-check-ratings
Politifact	Manual	Gaza-Israel, Ukraine-Russia	politifact.com/methodology
Misbar	API	Gaza-Israel, Ukraine-Russia	misbar.com/methodology
USA Today	API	Gaza-Israel, Ukraine-Russia	usatoday.com/fact-check-guidelines
Check Your Fact	API	Gaza-Israel, Ukraine-Russia	checkyourfact.com/methodology
Logically Facts	API	Gaza-Israel	logicallyfacts.com/faq
Newsmeter	API	Gaza-Israel	newsmeter.in/methodology
Newsweek	API	Ukraine-Russia	newsweek.com/ratings-fact-check
Newschecker	API	Ukraine-Russia	newschecker.in/home/our-methodology
Rappler	API	Ukraine-Russia	rappler.com/about

Table 16: An overview of the fact-checking websites which were used for the Gaza-Israel and/or Ukraine-Russia dataset and which have an existing methodology and label definitions. The table provides the website, the data collection type it was used in, the dataset(s) it was used for and an URL with its methodology.

Website	Data Collection Type	Dataset	Methodology URL
Newsmobile	API	Gaza-Israel, Ukraine-Russia	newsmobile.in/fact-check-methodology
Leadstories	API	Gaza-Israel, Ukraine-Russia	leadstories.com/how-we-work
Factchecker	API	Gaza-Israel, Ukraine-Russia	factchecker.gr/methodology
Factly	API	Gaza-Israel, Ukraine-Russia	factly.in/factchecking-methodology
The Quint	API	Gaza-Israel, Ukraine-Russia	thequint.com/about-us/webqoof
Voa News	API	Gaza-Israel, Ukraine-Russia	*No methodology article found*
Thip Media	API	Gaza-Israel, Ukraine-Russia	*No methodology article found*

Table 17: An overview of the fact-checking websites which were excluded from the Gaza-Israel and Ukraine-Russia dataset due to a lack of a) label definitions and/or b) an existing methodology. The table provides the website, the data collection type it was used in, the dataset(s) it was excluded from and (if available) an URL with its (unclear) methodology.

C.3 Original Label Definitions

Label	Definition
True	The knowledge from the fact-check supports the claim. The claim is factually accurate when it is confirmed by evidence from multiple and reliable sources. Mere plausibility is not enough for this decision.
False	The knowledge from the fact-check clearly refutes the claim. The claim is demonstrably false when it is disproven by evidence from multiple and reliable sources. The mere absence or lack of supporting evidence is not enough reason for being refuted.
Misleading	The claim or image is taken out of context, mispresented in a wrong context or necessary context is omitted. For example, an old claim or image is misrepresented in a new and different context. Other examples are to omit the context of a claim or image or to misinterpret and distort the meaning of a claim. In contrast to the label 'FALSE' the claim or image can also contain some true elements, which are taken out of context.
NEI	The fact-check does not contain sufficient information to come to a conclusion. For example,there is a substantial lack of evidence or the evidence is inconclusive, conflicting or self-contradictory. In the case of a lack of evidence, state which information exactly is missing. In particular, if no RESULTS or sources are available, pick this decision.

Table 18: The four labels and their definitions used in the preliminary DEFAME experiments on the Gaza-Israel and Ukraine-Russia datasets.

D Evaluation of DEFAME: Computational Costs, Evaluation Metrics and Results from Qualitative Analyses

D.1 Computational Costs

Model Variant	Time min/Claim	Model Calls	Input Tokens (Cost)	Output Tokens (Cost)	Total LLM API Cost	Google Calls	Google Vision Calls	Total Search Calls
DEFAME	1:46	1213	6.6M (0.49€)	117K (0.04€)	0.53€	156	55	211
w/o Web Search	0:36	1280	8.5M (0.64€)	98K (0.03€)	0.67€	168	59	227
w/o Reverse Image Search	1:38	1200	7.2M (0.54€)	111K (0.03€)	0.57€	193	0	193
w/o Image Search	3:02	1169	5.2M (0.39€)	118K (0.04€)	0.43€	142	59	201
w/o Geolocation	1:27	1184	6.1M (0.46€)	115K (0.03€)	0.49€	162	53	215

Table 19: An overview of the use and computational cost statistics for the different DEFAME variants on the Gaza-Israel dataset. The statistics are for one replication run. The column google calls denotes the number of calls for web and image search.

Model Variant	Time min/Claim	Model Calls	Input Tokens (Cost)	Output Tokens (Cost)	Total LLM API Cost	Google Calls	Google Vision Calls	Total Search Calls
DEFAME	1:15	848	4.0M (0.30€)	89K (0.03€)	0.32€	119	17	136
w/o Web Search	0:56	1190	12M (0.88€)	74K (0.02€)	0.91€	173	24	197
w/o Reverse Image Search	1:18	953	4.9M (0.37€)	93K (0.03€)	0.40€	155	0	155
w/o Image Search	1:27	849	4.4M (0.33€)	88K (0.03€)	0.36€	119	21	140
w/o Geolocation	1:14	843	4.0M (0.30€)	88K (0.03€)	0.33€	124	14	138

Table 20: An overview of the use and computational cost statistics for the different DEFAME variants on the Ukraine-Russia dataset. The statistics are for one replication run. Google calls denotes the number of calls for web and image search.

D.2 Evaluation Metrics

Metric	Notation	Formula	Range
Balanced Accuracy	BA	$\frac{\frac{TP}{TP+FN}+\frac{TN}{TN+FP}}{2}$	[0;1]
Matthews Correlation Coefficient	MCC	$\frac{TP\times TN-FP\times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	[-1;1]
True Negative Rate/Specificity	TNR	$\frac{TN}{TN+FP}$	[0;1]
Macro-Averaged F_1	Macro F_1	$\frac{F_{1\text{True}}+F_{1\text{False}}}{2}$	[0;1]
Number of Correct Predictions	Correct *(n)*	—	[0;n]
Number of Wrong Predictions	Wrong *(n)*	—	[0;n]
Number of Wrong Predictions across all three replication runs	Consistently Wrong *(n)*	—	[0;n]
Number of False Positives	FP *(n)*	—	[0;94]
Number of False Negatives	FN *(n)*	—	[0;9]

Table 21: An overview of the evaluation metrics used on the dataset-level for the Gaza-Israel and Ukraine-Russia dataset. The table shows the name, notation, mathematical formula and the range of values of a metric. The mathematical formulas are taken from [28, 54]. n represents the size of each dataset (79 for the Ukraine-Russia dataset, 100 for the Gaza-Israel dataset).

Metric	Notation	Formula	Range
Balanced Accuracy	BA	$\frac{\frac{TP}{TP+FN}+\frac{TN}{TN+FP}}{2}$	[0;1]
Matthews Correlation Coefficient	MCC	$\frac{TP\times TN-FP\times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}}$	[-1;1]
True Negative Rate/Specificity	TNR	$\frac{TN}{TN+FP}$	[0;1]
Precision for *True* label	Precision (True)	$\frac{TP}{TP+FP}$	[0;1]
Number of Correct Predictions	Correct *(n)*	—	[0;n]
Number of Wrong Predictions	Wrong *(n)*	—	[0;n]
Number of Wrong Predictions across all three replication runs	Consistently Wrong *(n)*	—	[0;n]
Number of False Positives	FP *(n)*	—	[0;94]
Number of False Negatives	FN *(n)*	—	[0;9]

Table 22: An overview of the evaluation metrics used on the claim type-level for the Gaza-Israel and Ukraine-Russia dataset. The table shows the name, notation, mathematical formula and the range of values of a metric. The mathematical formulas are taken from [28, 54]. n represents the size of each dataset (79 for the Ukraine-Russia dataset, 100 for the Gaza-Israel dataset). Note that the computation of Matthews Correlation Coefficient (MCC) requires True Positives (TP) and False Positives (FP). Since all claims with AI or altered images have the label *False*, this metric cannot be computed for these claim types. Thus, it will not be shown for the respective claim types in Table 8 and Table 10.

D.3 Results from Qualitative Analysis

Claim Index	Claim	Claim Type	DEFAME variant(s)	Frequency
26	This image is an authentic screenshot of J.D. Vance's X profile that says 'I stand with Israel'.	Altered Image	DEFAME (default), w/o Web Search, w/o Reverse Image Search, w/o Geolocate	14
89	For the assassination on July 31, 2024, Israeli intelligence tracked Hamas leader Ismail Haniyeh through WhatsApp.	Text-Only	w/o Reverse Image Search	10
70	The Iron Dome intercepted 90% of ballistic missiles fired at Israel in Iran's attack on October 1, 2024.	Text-Only	w/o Reverse Image Search	9
91	This image shows the aftermath of a Houthi attack on Tel Aviv, Israel, in July 2024.	Normal Image	w/o Web Search	9
77	Israel experienced an earthquake on October 5, 2024.	Text-Only	w/o Reverse Image Search	9
9	This image shows an Israeli missile attack on Iran in the early hours of October 26, 2024.	Normal Image	w/o Geolocate	8
90	An Algerian judo athlete withdrew from his Olympics 2024 match against an Israeli opponent.	Text-Only	DEFAME (default)	8
8	This image shows Israel launching several waves of revengeful attacks on Iranian military targets on October 26, 2024.	Normal Image	w/o Reverse Image Search	7
17	Attack on a US-flagged vessel in the Gulf of Aden ...	Text-Only	w/o Image Search	7
34	This image shows Israel bombing Tehran on October 26, 2024.	Normal Image	w/o Reverse Image Search	5

Table 23: The 10 claims from the Gaza-Israel dataset that were consistently wrongly predicted within one or several DEFAME variants across all three replication runs. The table shows the claim index, claim, claim type, the DEFAME variant(s) within which it was consistently wrongly predicted, and the overall frequency of being wrongly predicted across all DEFAME variants and replication runs. *Note*: The claim with the index 17 is not fully shown, because it is too long.

Claim Index	Claim Type	Frequency
26	Altered Image	14
89	Text-Only	10
70	Text-Only	9
91	Normal Image	9
77	Text-Only	9
9	Normal Image	8
90	Text-Only	8
8	Normal Image	7
58	Normal Image	7
17	Text-Only	7

Table 24: The 10 claims from the Gaza-Israel dataset with the highest frequency of being wrongly predicted across all DEFAME variants and replication runs. The table provides their index, claim type and frequency.

Claim Index	Claim	Claim Type	DEFAME variant	Frequency
31	In August 2024, Ukraine passed legislation banning the Orthodox Church in Ukraine.	Text-Only	w/o Web Search, w/o Reverse Image Search, w/o Geolocate, w/o Image Search	14
63	In an interview with NDTV on March 17, 2025 Director of US National Intelligence Tulsi Gabbard said that President Donald Trump and Russian President Vladimir Putin were good friends.	Text-Only	w/o Reverse Image Search, w/o Geolocate	12
48	This image shows an authentic transcript of the heated White House meeting between US president Donald Trump, vice-president JD Vance and Ukrainian president Volodymyr Zelenskyy on February 28, 2025.	Altered Image	w/o Reverse Image Search, w/o Geolocate	11
28	In March 2025 former President of Poland Lech Wałęsa wrote a letter to Donald Trump criticizing the U.S. President's decision to suspend the delivery of all U.S. military aid to Ukraine.	Text-Only	w/o Web Search	10
18	In November 2024 Russia sentenced a man to 14 years in prison for burning the Quran.	Text-Only	w/o Web Search	9
32	U.S. Vice President Kamala Harris once said, 'Ukraine is a country in Europe...'	Text-Only	w/o Web Search	8
25	U.S. Vice President JD Vance's cousin Nate Vance fought in Ukraine for three years during its war with Russia.	Text-Only	w/o Web Search	3

Table 25: The 7 claims from the Ukraine-Russia dataset that were consistently wrongly predicted within one or several DEFAME variants across all three replication runs. The table shows the claim index, claim, claim type, the DEFAME variant(s) within it was consistently wrongly predicted and the overall frequency of being wrongly predicted across all DEFAME variants and replication runs. *Note*: The claim with the index 32 is not fully shown, because it is too long.

Claim Index	Claim Type	Frequency
31	Text-Only	14
63	Text-Only	12
48	Altered Image	11
28	Text-Only	10
18	Text-Only	9
32	Text-Only	8
71	Text-Only	8
16	Text-Only	7
37	Text-Only	7
75	Normal Image	4

Table 26: The 10 claims from the Ukraine-Russia dataset with the highest frequency of being wrongly predicted across all DEFAME variants and replication runs. The table provides their index, claim type and frequency.

Claim Index	Claim Type	Consistently Wrong within DEFAME variant	Error Frequency	Primary Error Reason	Secondary Error Reason	Tool Limitation	Evidence Quality
26	Altered Image	w/o RIS, default, w/o web search, w/o geolocate	14	Failed retrieval/ understanding of sources, Context/retrieved evidence makes claim highly plausible	Model does not verify image content or Wrong RIS results	No web search available or Relies on contextual understanding only	Poor to moderate - varies by available tools
89	Text-Only	w/o RIS	10	Context/retrieved evidence makes claim highly plausible	Ground truth label may not be rigorous and might be wrong	Difficult to verify assassination details	Contradictory sources (misbar vs intellinews)
70	Text-Only	w/o RIS	9	Context/retrieved evidence makes claim highly plausible	Linguistic/contextual misunderstanding: Mixed up general vs specific success rates	-	Conflicting evidence in sources
91	Normal Image	w/o web search	9	Context/retrieved evidence makes claim highly plausible	News sources falsely use image from different context	No web search for broader context verification	Misleading - correct geolocation but wrong temporal context
77	Text-Only	w/o RIS	9	Wrong retrieval/ understanding of sources	Misinformation in NDTV article	-	Conflicting evidence in sources
9	Normal Image	w/o geolocate	8	News sources contain wrong information: Old image used in new context	-	No geolocation to verify image origin	Misleading - newspapers wrongly use old images
90	Text-Only	default	8	Claim ambiguity	Interpretation dependent on intentionality	-	Ambiguous evidence interpretation
8	Normal Image	w/o RIS	7	Context/retrieved evidence make claim highly plausible	Similar image context (Israel bombing in Lebanon vs Iran)	No RIS to identify exact image origin	High plausibility from web/image search
17	Text-Only	w/o image search	7	Ground truth label error	Missing image makes claim actually true	Missing image content	Predictions actually correct
34	Normal Image	w/o RIS	5	Context/retrieved evidence makes claim highly plausible	CNN article image reference could not be verified	-	Web evidence highly plausible but mentioned image could not be found

Table 27: Results from the qualitative error analysis on the Gaza-Israel dataset of the claims with consistently wrong predictions across all three replication runs within each DEFAME variant. The table shows the claim index and type, the DEFAME variant, the overall error frequency of each claim across all DEFAME variants and all replication runs and the results from the qualitative analysis: the primary and secondary error reasons and notes about tool limitations and evidence quality.

Claim Index	Claim Type	Consistently Wrong within DEFAME variant	Error Frequency	Primary Error Reason	Secondary Error Reason	Tool Limitation	Evidence Quality
31	Text-Only	w/o RIS, w/o web search, w/o geolocate, w/o image search	14	News sources contain misleading/wrong headlines and/or content: Orthodox Church of Ukraine vs Ukrainian Orthodox Church.	-	-	Misleading
63	Text-Only	w/o RIS, w/o geolocate	12	News sources contain factually wrong information	-	-	Factually wrong information
48	Altered Image	w/o RIS, w/o geolocate	11	The model did not check or compare the actual content of the Altered image with the retrieved evidence.	The context of the retrieved evidence makes the claim highly plausible.	-	-
28	Text-Only	w/o web search	10	Claim ambiguity. Mismatch between fact-checking website evidence and retrieved newspaper evidence.	Snopes based its fact-check on criticisms around military aid, the retrieved newspapers confirm the criticism but not the military aid part.	-	-
18	Text-Only	w/o web search	9	News sources contain wrong/misleading/ ambiguous headlines and/or information: Quran Burning vs Treason.	-	-	Misleading
32	Text-Only	w/o web search	8	Tool limitation: Could not retrieve enough relevant evidence without web search to make the correct prediction.	-	Could not retrieve enough relevant evidence without web search.	-
25	Text-Only	w/o web search	3	Tool limitation: Could not retrieve enough relevant evidence without web search to make the correct prediction.	-	Could not retrieve enough relevant evidence without web search.	-

Table 28: Results from the qualitative error analysis on the Ukraine-Russia dataset of the claims with consistently wrong predictions across all three replication runs within each DEFAME variant. The table shows the claim index and type, the DEFAME variant, the overall error frequency of each claim across all DEFAME variants and all replication runs and the results from the qualitative analysis: the primary and secondary error reasons and notes about tool limitations and evidence quality.